US012412841B2

(12) **United States Patent**
Tsai et al.

(10) **Patent No.: US 12,412,841 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **SEMICONDUCTOR PACKAGE AND MANUFACTURING METHOD THEREOF**

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(73) Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 56 days.

(21) Appl. No.: **18/523,895**

(22) Filed: **Nov. 30, 2023**

(65) **Prior Publication Data**

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Related U.S. Application Data

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(Continued)

(51) **Int. Cl.**
H01L 23/538 (2006.01)
H01L 21/48 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***H01L 23/5383*** (2013.01); ***H01L 21/4857*** (2013.01); ***H01L 21/486*** (2013.01); ***H01L 21/565*** (2013.01); ***H01L 23/3121*** (2013.01); ***H01L 23/49822*** (2013.01); ***H01L 23/5226*** (2013.01); ***H01L 24/05*** (2013.01); ***H01L 25/0655*** (2013.01); ***H01L 25/50*** (2013.01); *H01L 2224/02372* (2013.01)

(58) **Field of Classification Search**
CPC H01L 25/0655; H01L 23/5383; H01L 23/49822; H01L 23/5226; H01L 21/4857; H01L 21/486; H01L 2224/02372; H01L 25/105; H01L 25/0657
See application file for complete search history.

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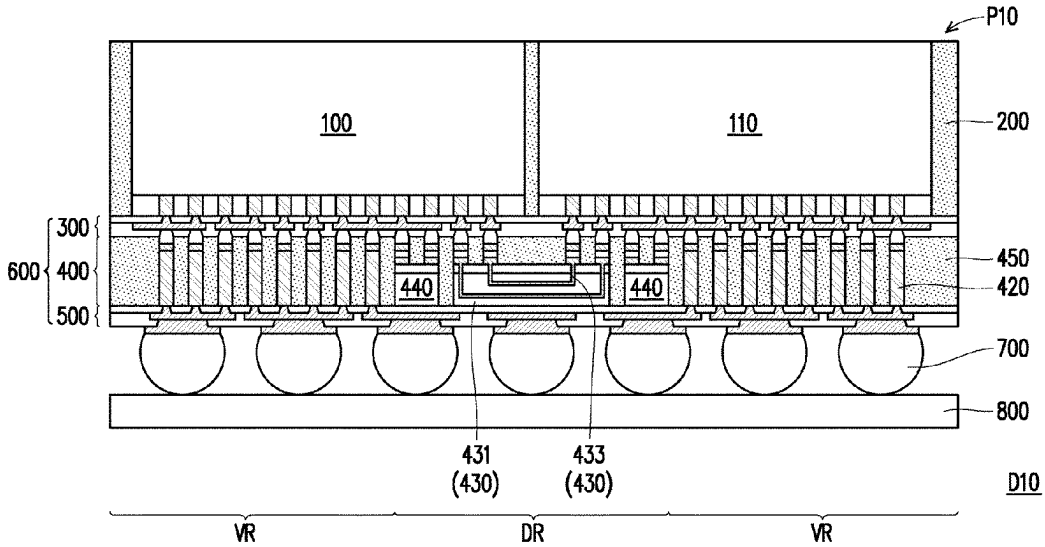
Primary Examiner — Antonio B Crite
(74) *Attorney, Agent, or Firm* — JCIPRNET

(57) **ABSTRACT**

A semiconductor package includes a first semiconductor die, a second semiconductor die, a semiconductor bridge, an integrated passive device, a first redistribution layer, and connective terminals. The second semiconductor die is disposed beside the first semiconductor die. The semiconductor bridge electrically connects the first semiconductor die with the second semiconductor die. The integrated passive device is electrically connected to the first semiconductor die. The first redistribution layer is disposed over the semiconductor bridge. The connective terminals are disposed on the first redistribution layer, on an opposite side with respect to the semiconductor bridge. The first redistribution layer is interposed between the integrated passive device and the connective terminals.

20 Claims, 36 Drawing Sheets



Related U.S. Application Data

(60) Provisional application No. 62/905,431, filed on Sep. 25, 2019.

(51) **Int. Cl.**

H01L 21/56	(2006.01)
H01L 23/00	(2006.01)
H01L 23/31	(2006.01)
H01L 23/498	(2006.01)
H01L 23/522	(2006.01)
H01L 25/00	(2006.01)
H01L 25/065	(2023.01)
H01L 25/10	(2006.01)
H01L 25/16	(2023.01)
H01L 21/683	(2006.01)

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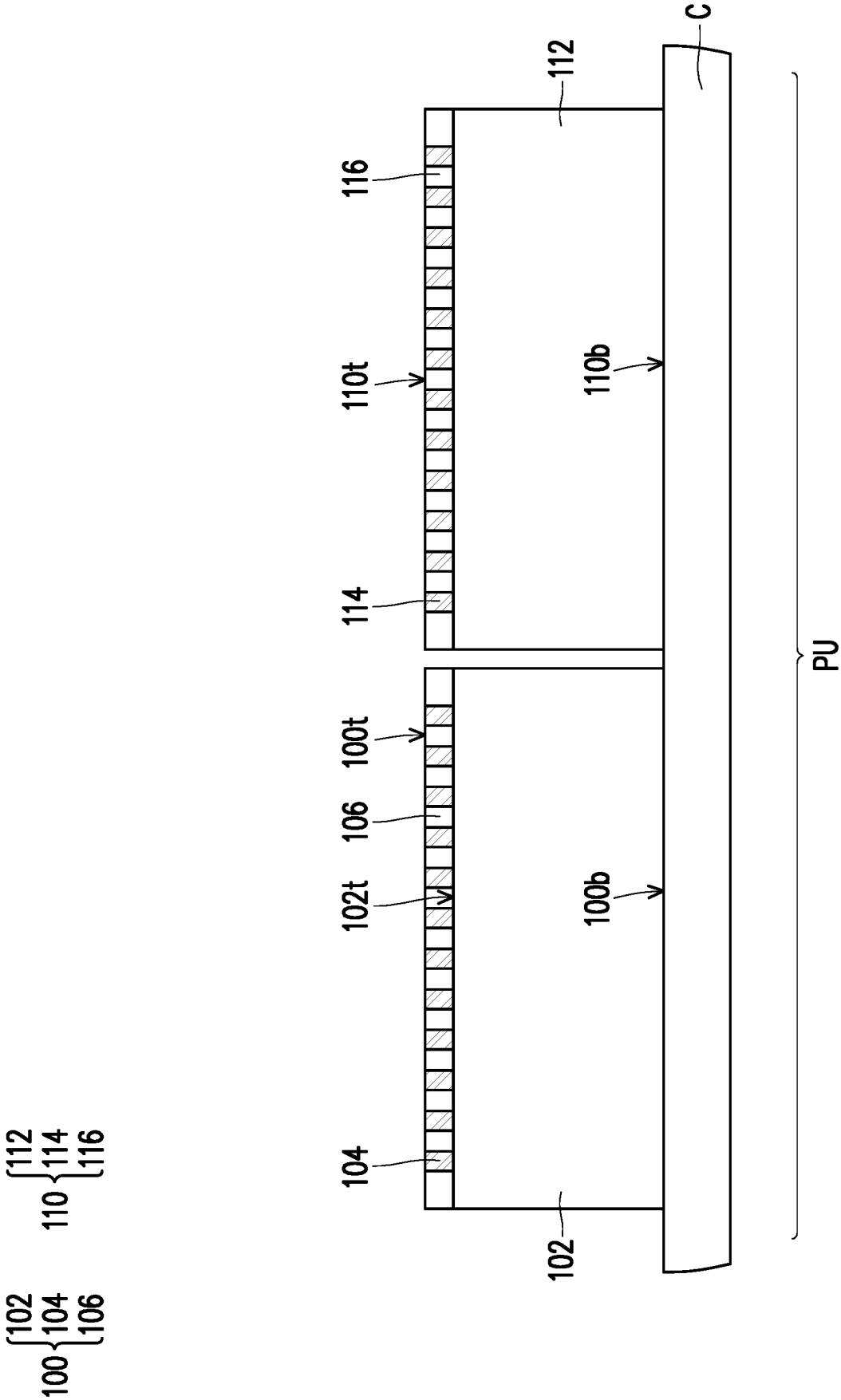
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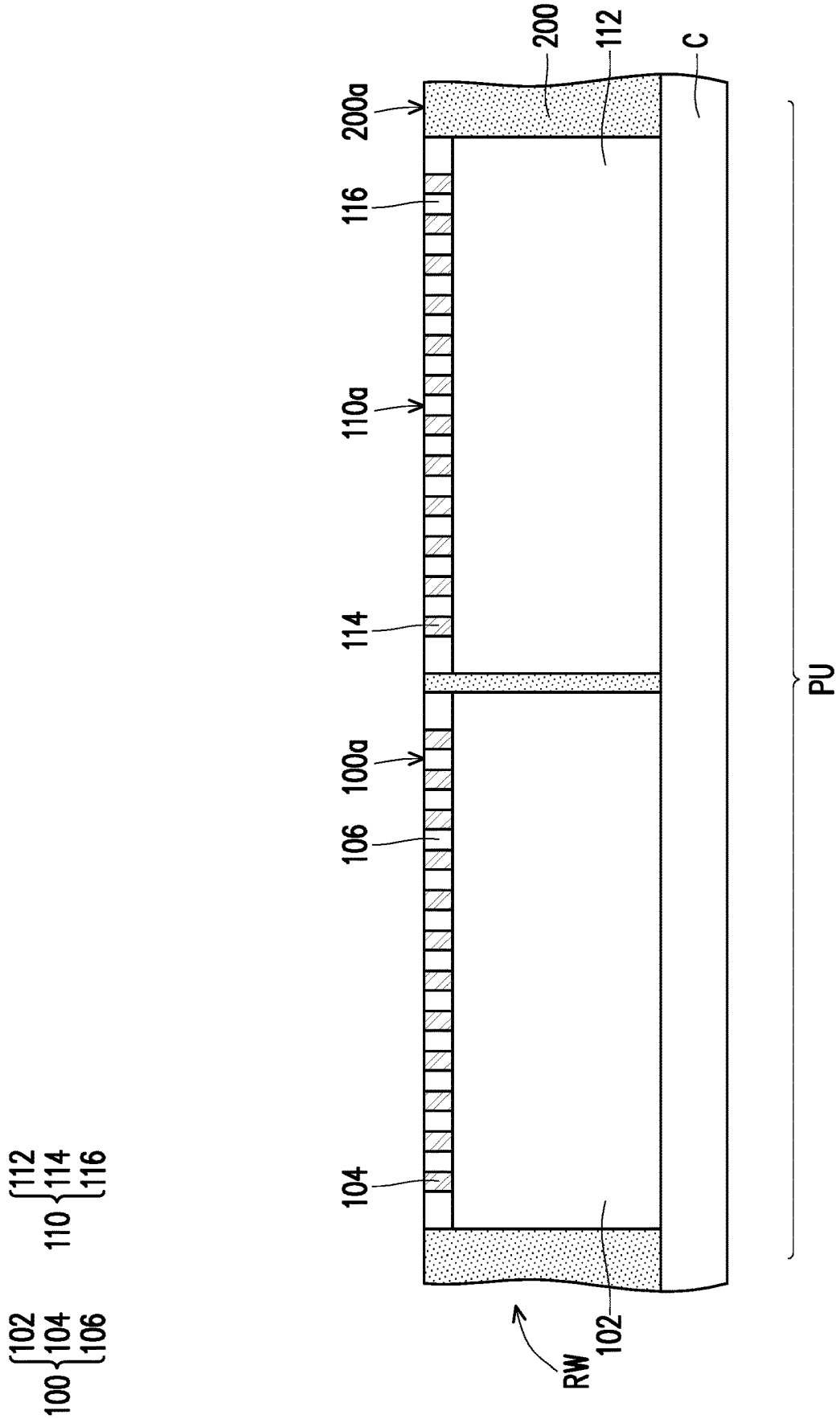
102
104
106
110
112
114
116
102
104
102t
106
100t
114
110t
116
112
100b
110b
C
PU

FIG. 1A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112
C
200a
116
110a
114
PU
100a
106
104
102
RW
112
114
116
110
102
104
106
100

FIG. 1B

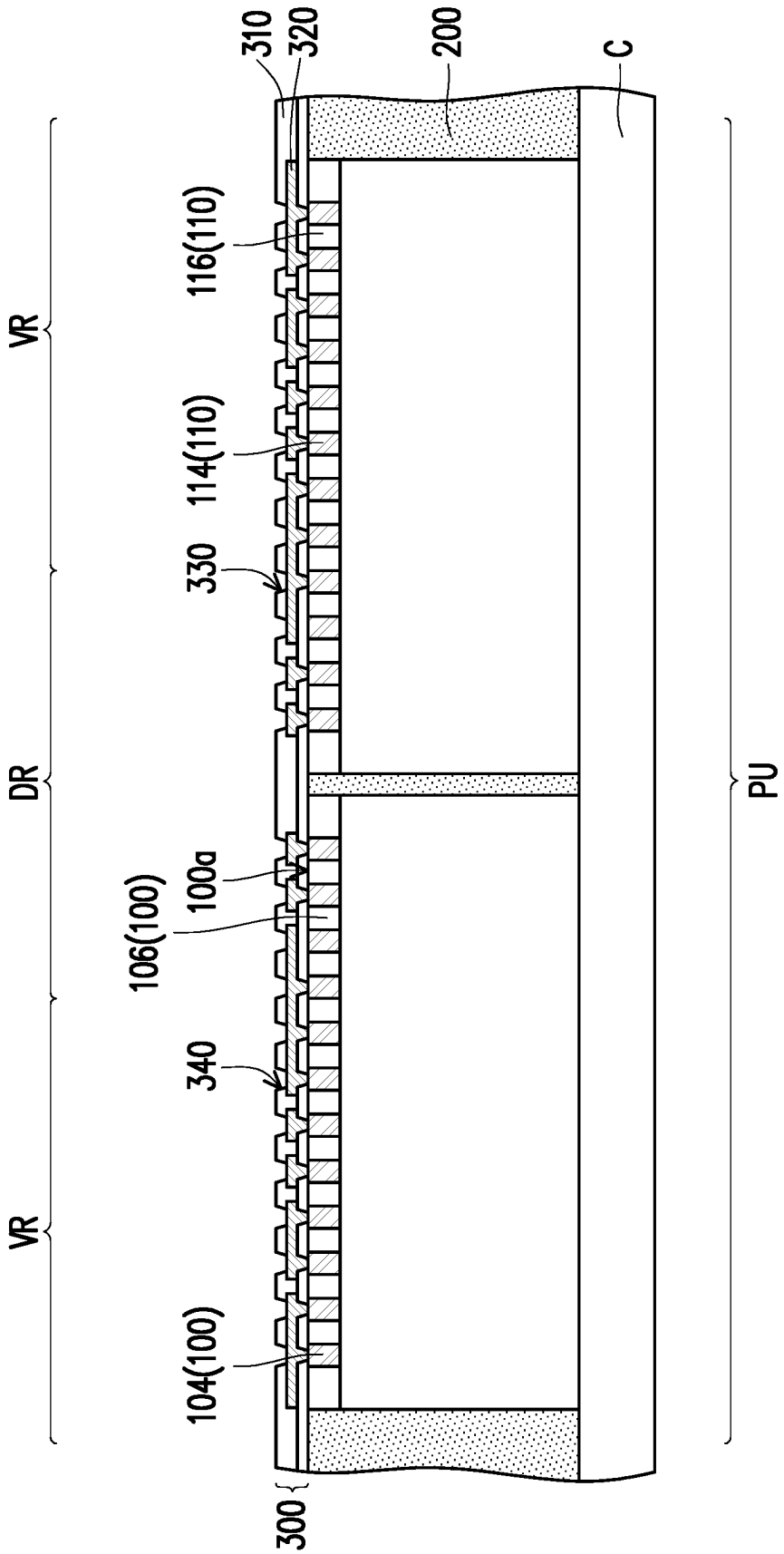

310
320
200
C
116(110)
114(110)
330
100a
106(100)
340
104(100)
300
VR
DR
VR
PU

FIG. 1C

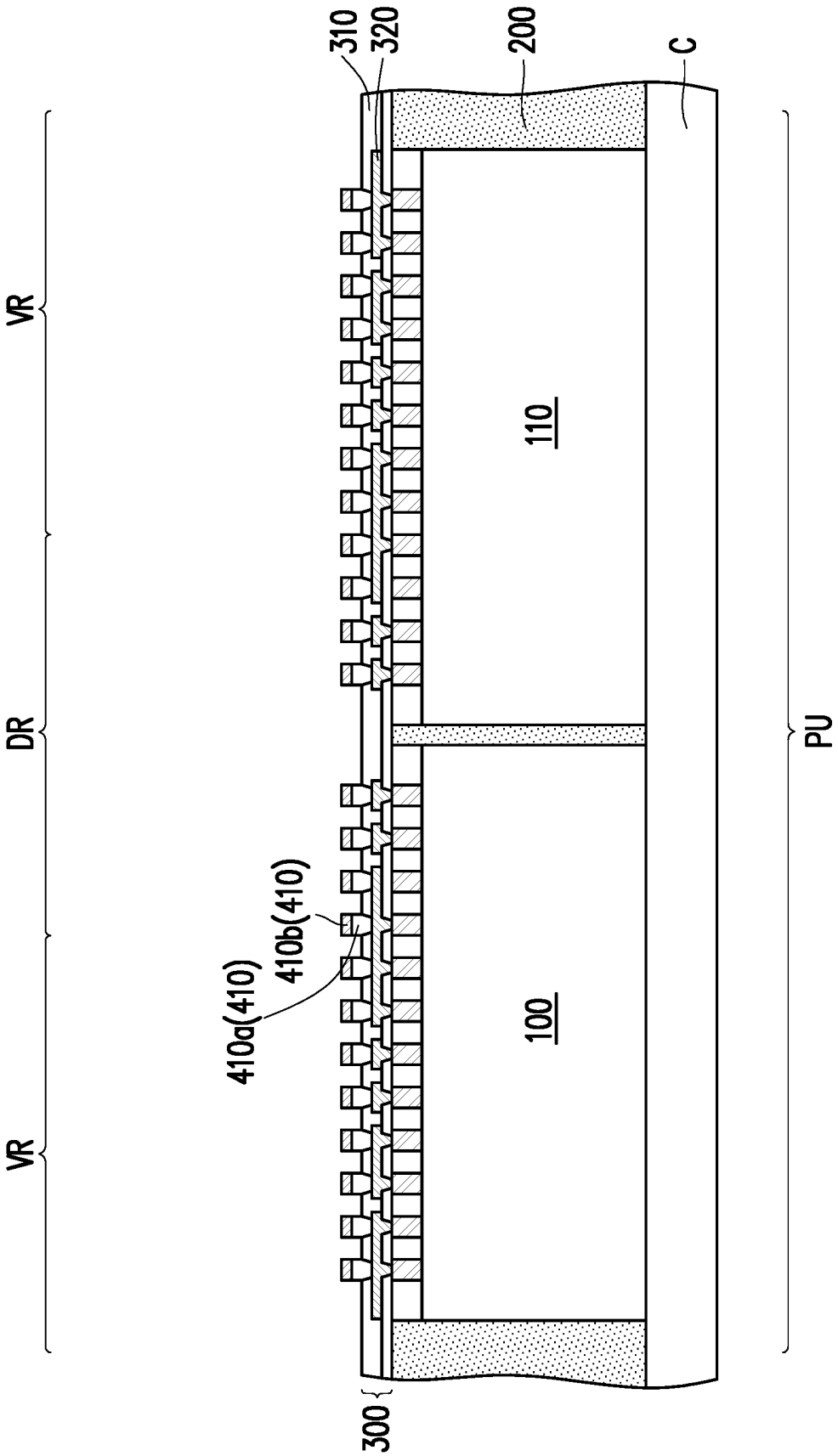


FIG. 1D

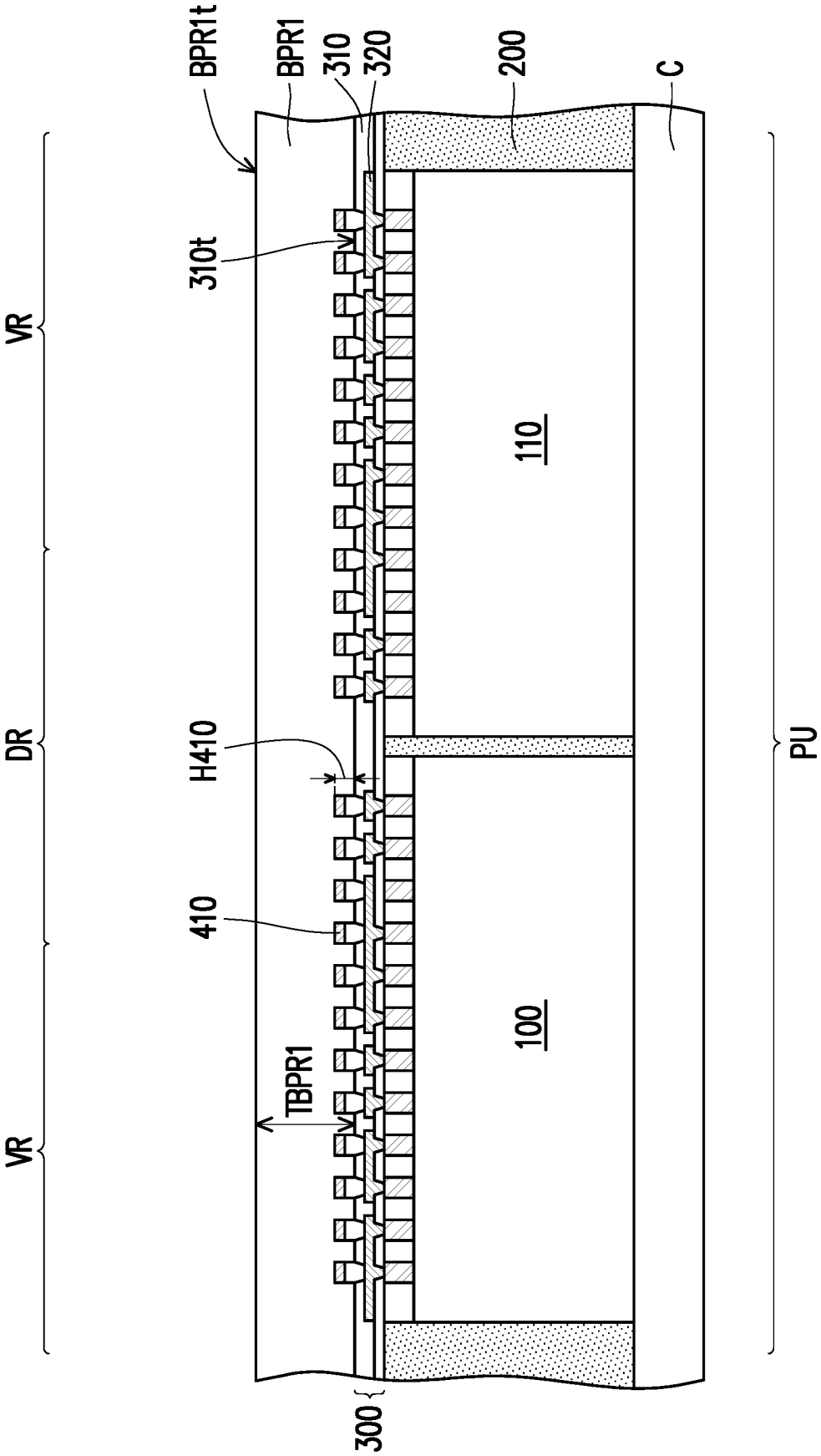

BPR1t
BPR1
310
320
200
C
310t
VR
DR
VR
110
H410
410
100
TBPR1
PU
300

FIG. 1E

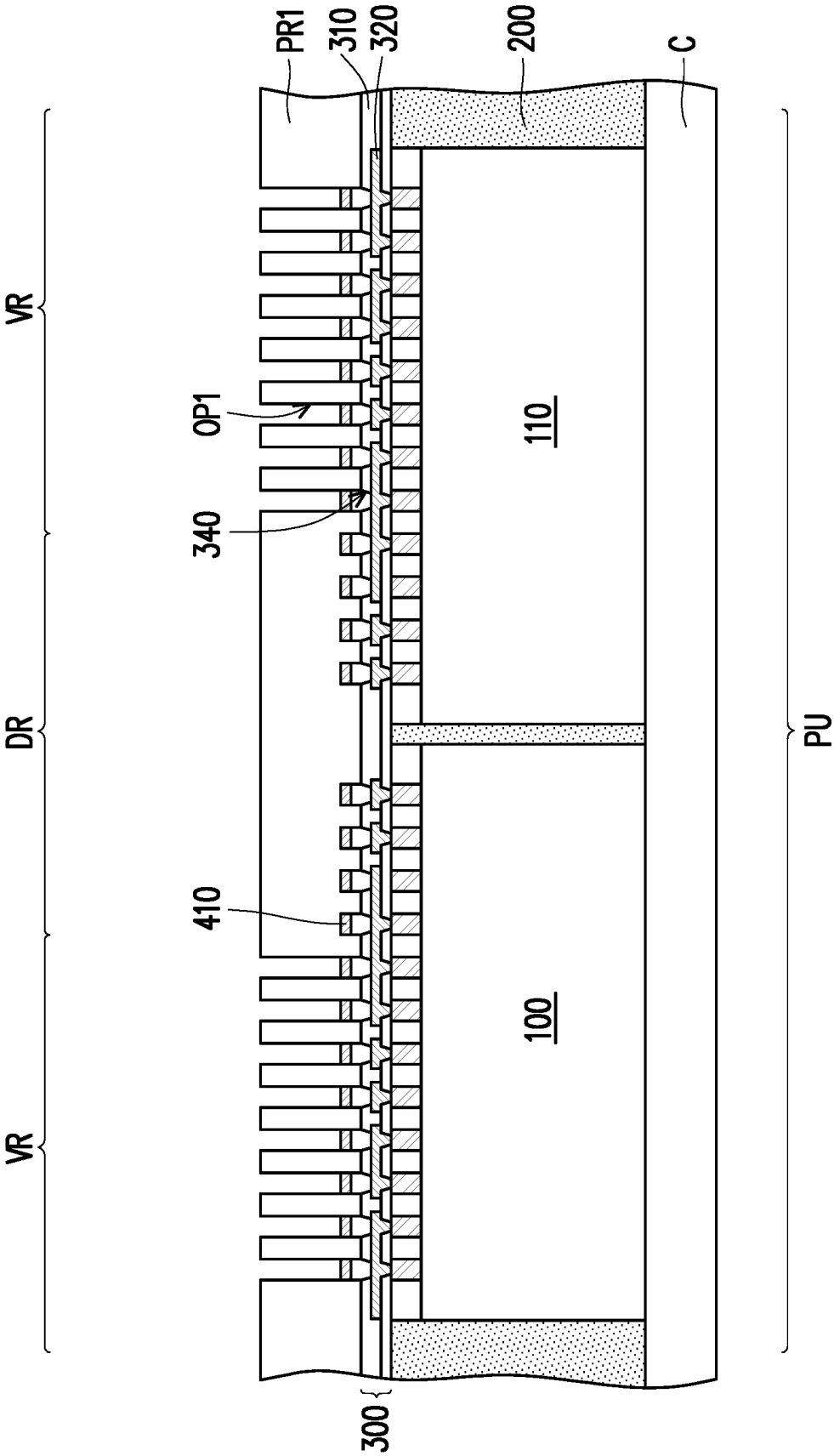


FIG. 1F

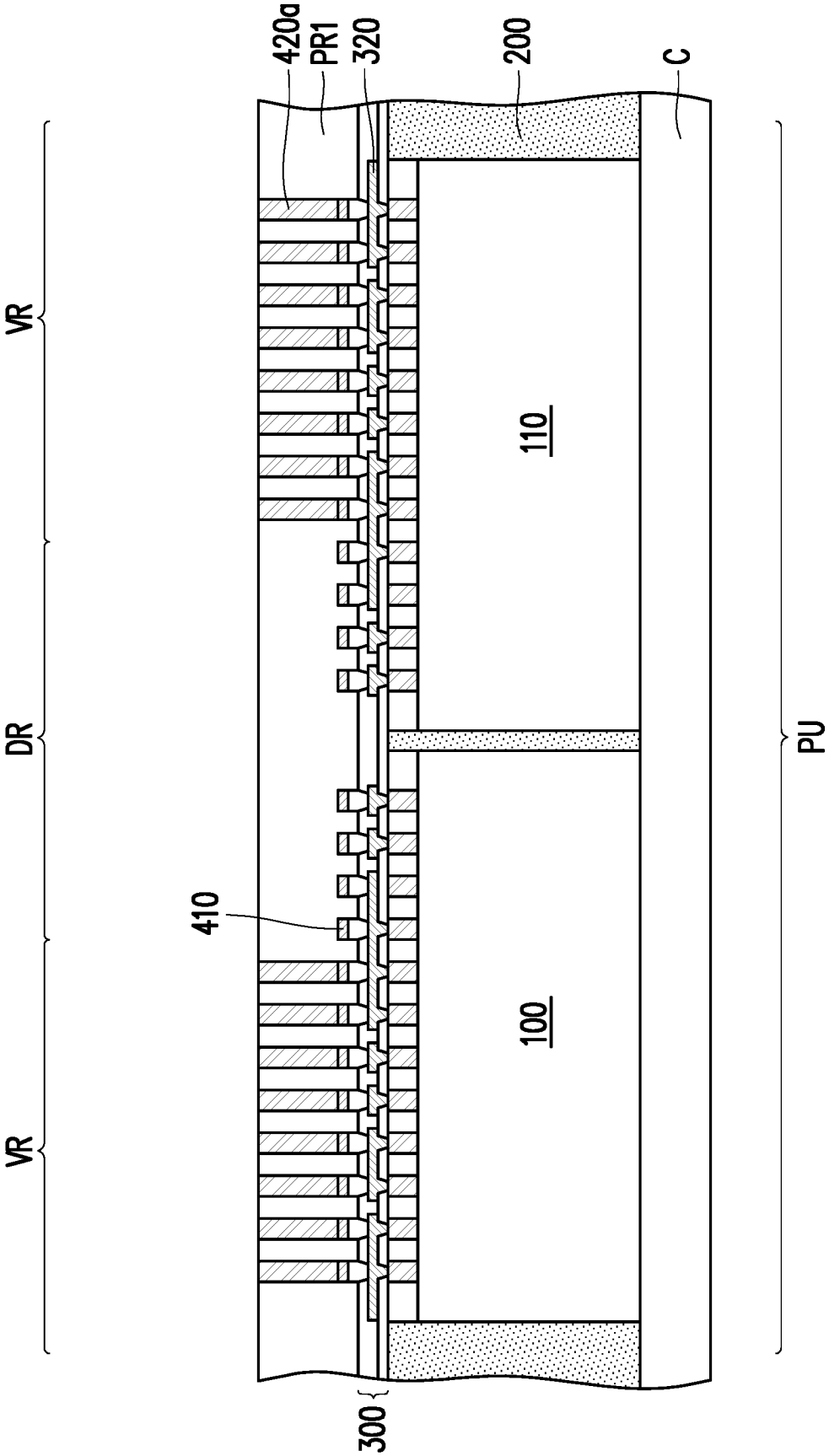


FIG. 1G

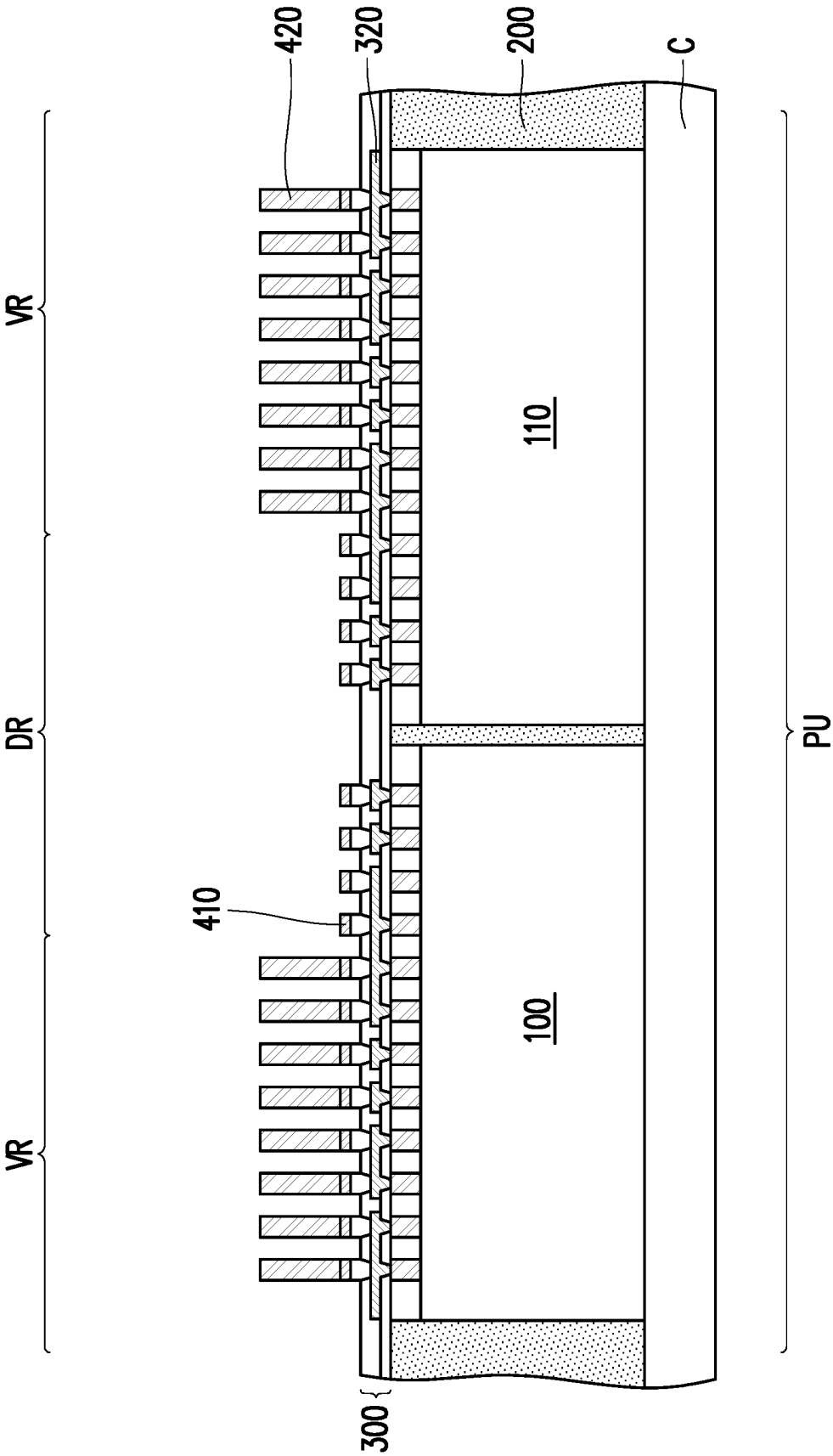


FIG. 1H

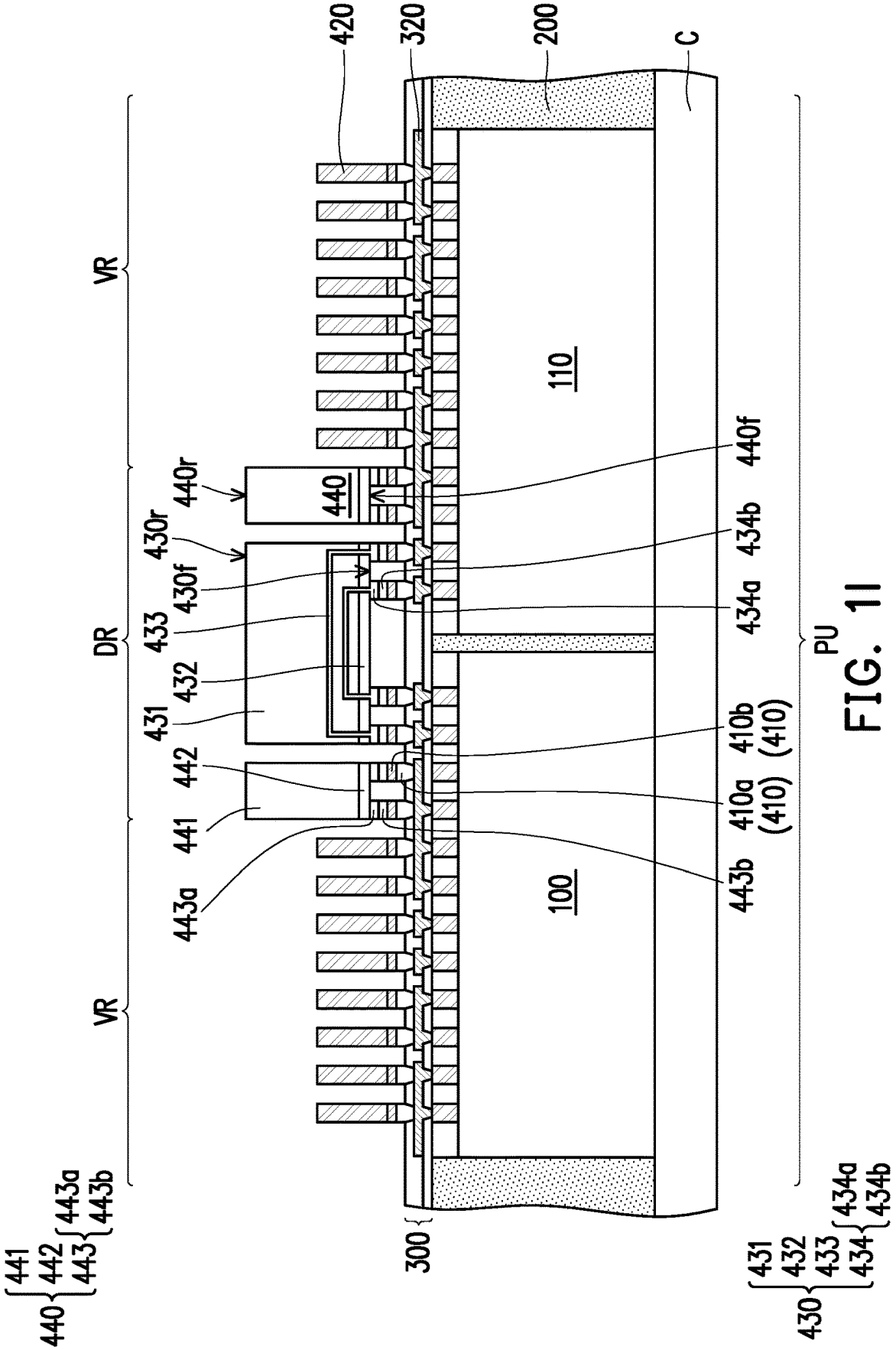

420
320
200
C
VR
110
440f
440r
440
430r
434b
430f
434a
433
DR
432
431
410b
(410)
442
410a
(410)
441
443b
443a
100
PU
VR
443a
443b
441
442
443
440
300
431
432
433
434
430
434a
434b

FIG. 11

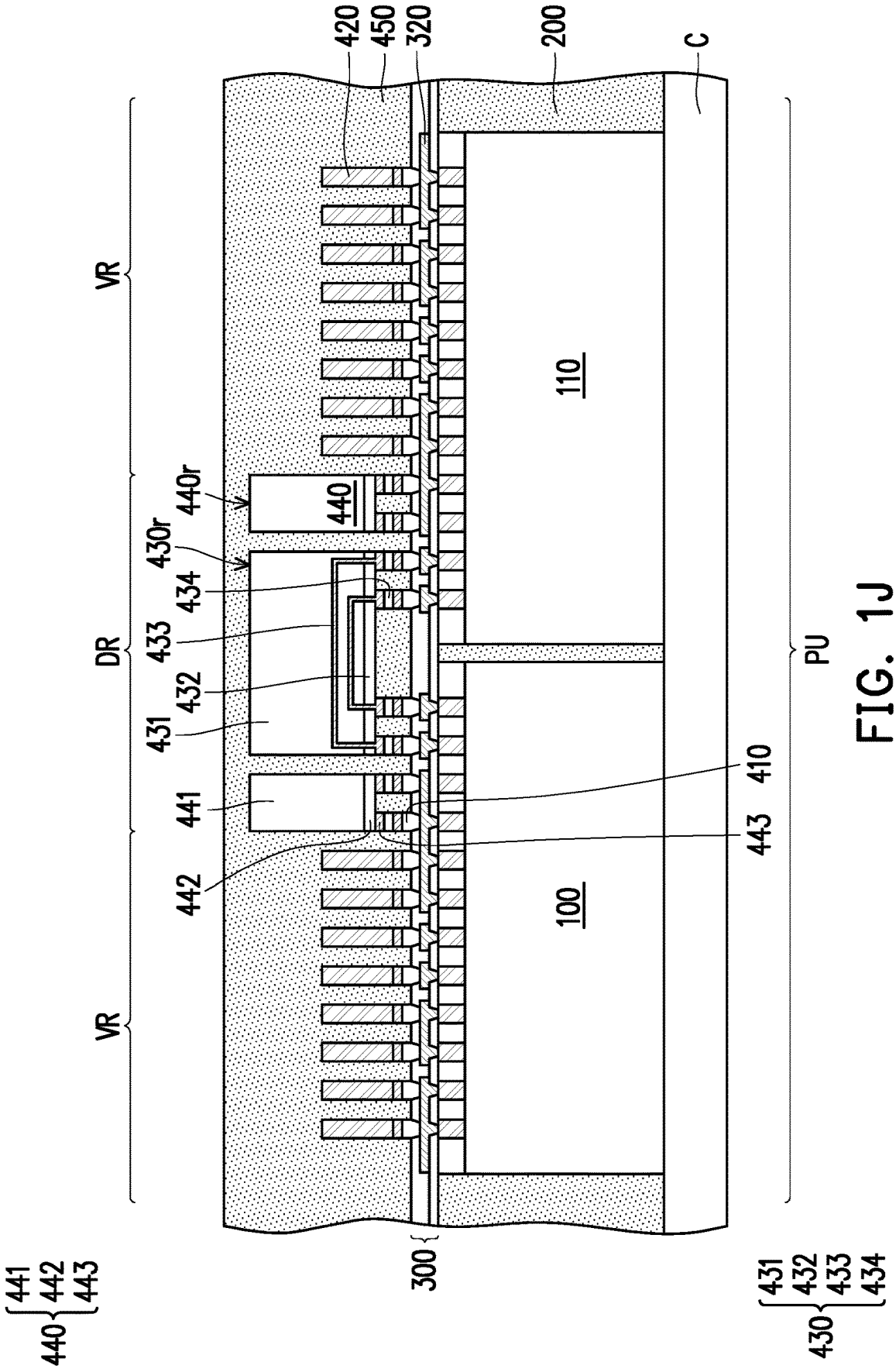

420
450
320
200
C
VR
DR
VR
PU
110
100
440
440r
430r
434
433
432
431
441
442
410
443
300
441
442
443
440
431
432
433
434
430

FIG. 1J

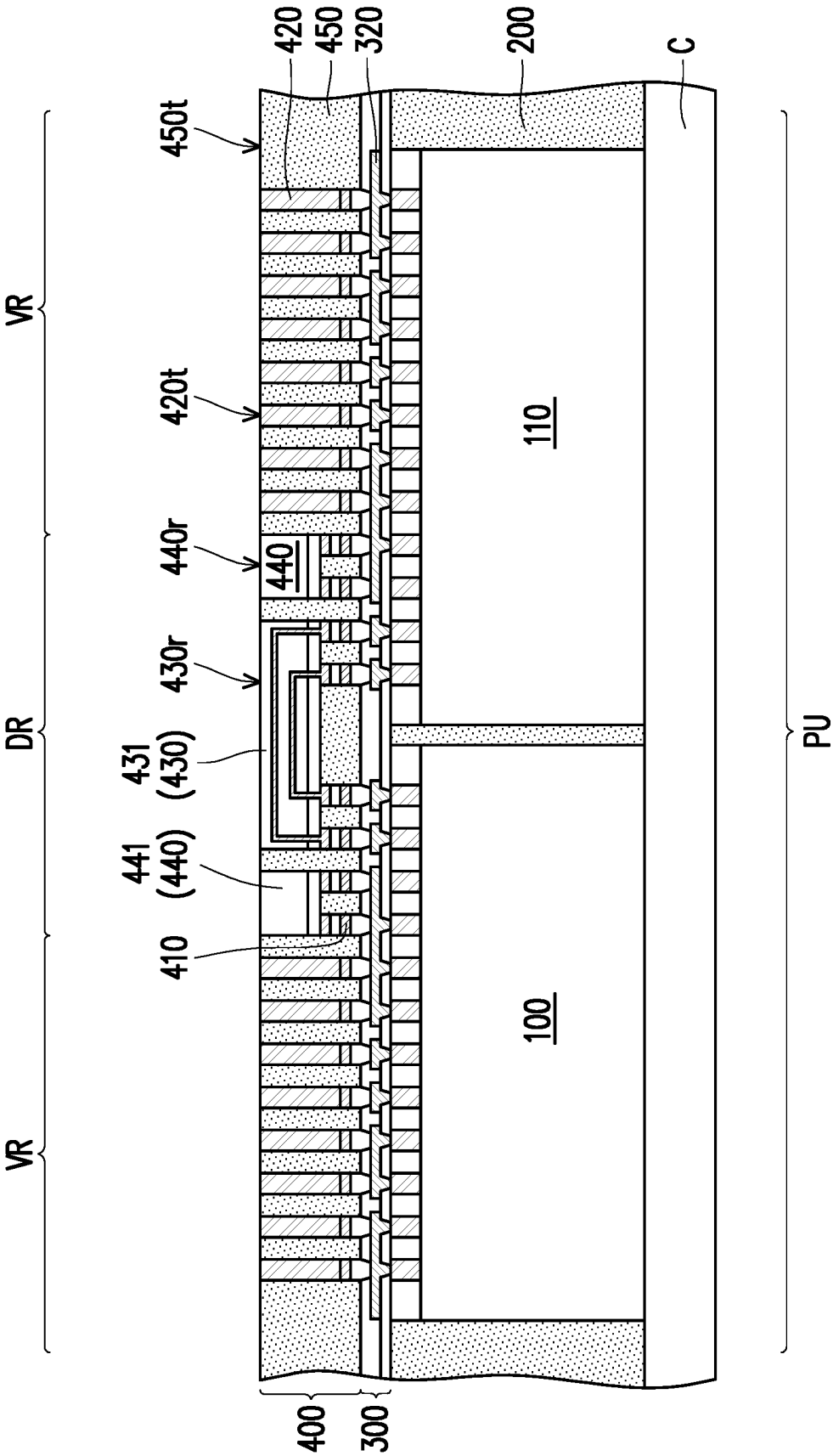

420
450
320
200
C
450t
VR
420t
110
440r
440
430r
DR
431
(430)
441
(440)
410
100
VR
PU
400
300

FIG. 1K

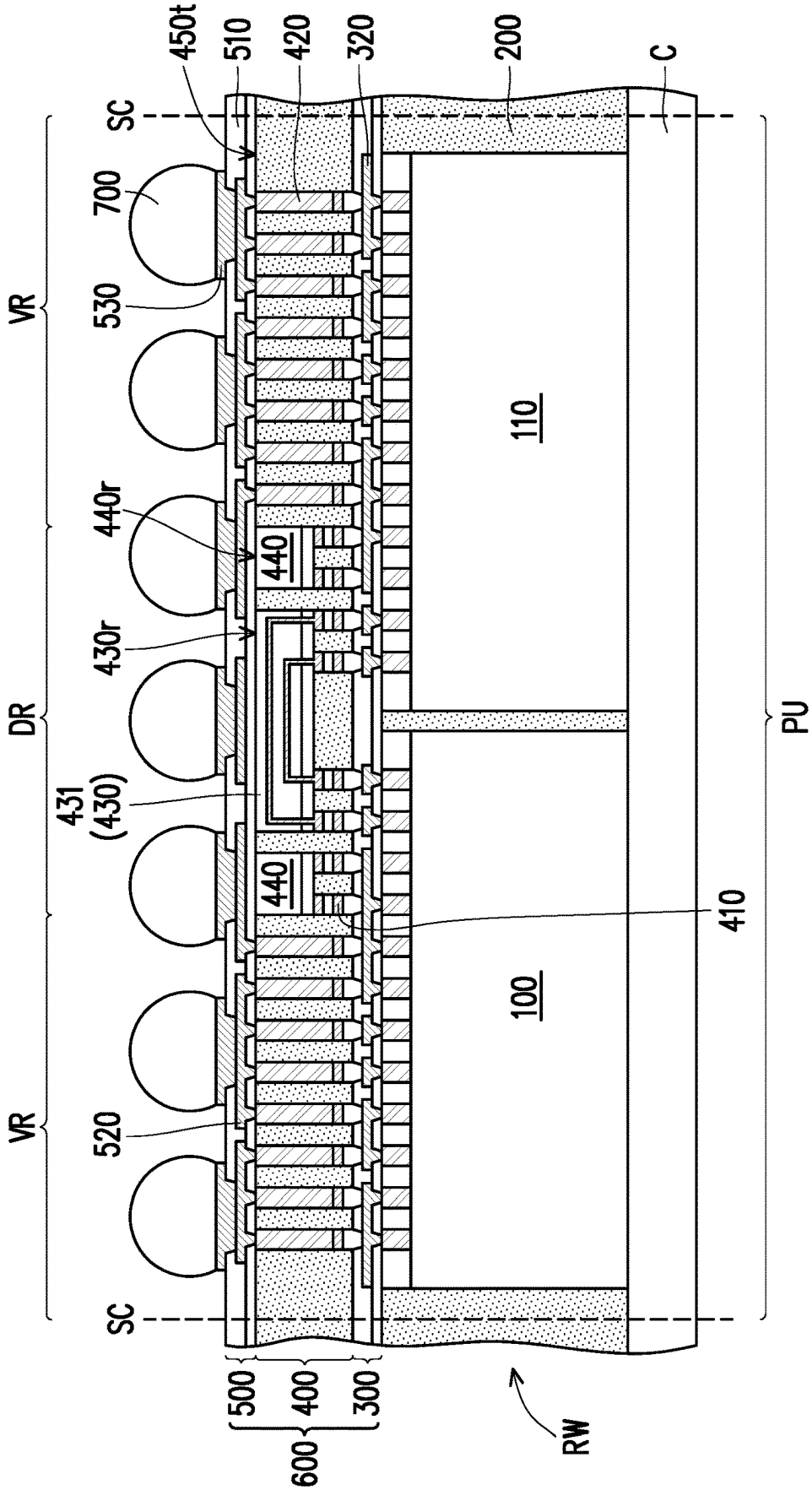

450t
510
420
320
200
C
SC
700
530
VR
440r
440
430r
DR
431
(430)
440
410
110
100
PU
520
VR
SC
500
400
300
600
RW

FIG. 1L

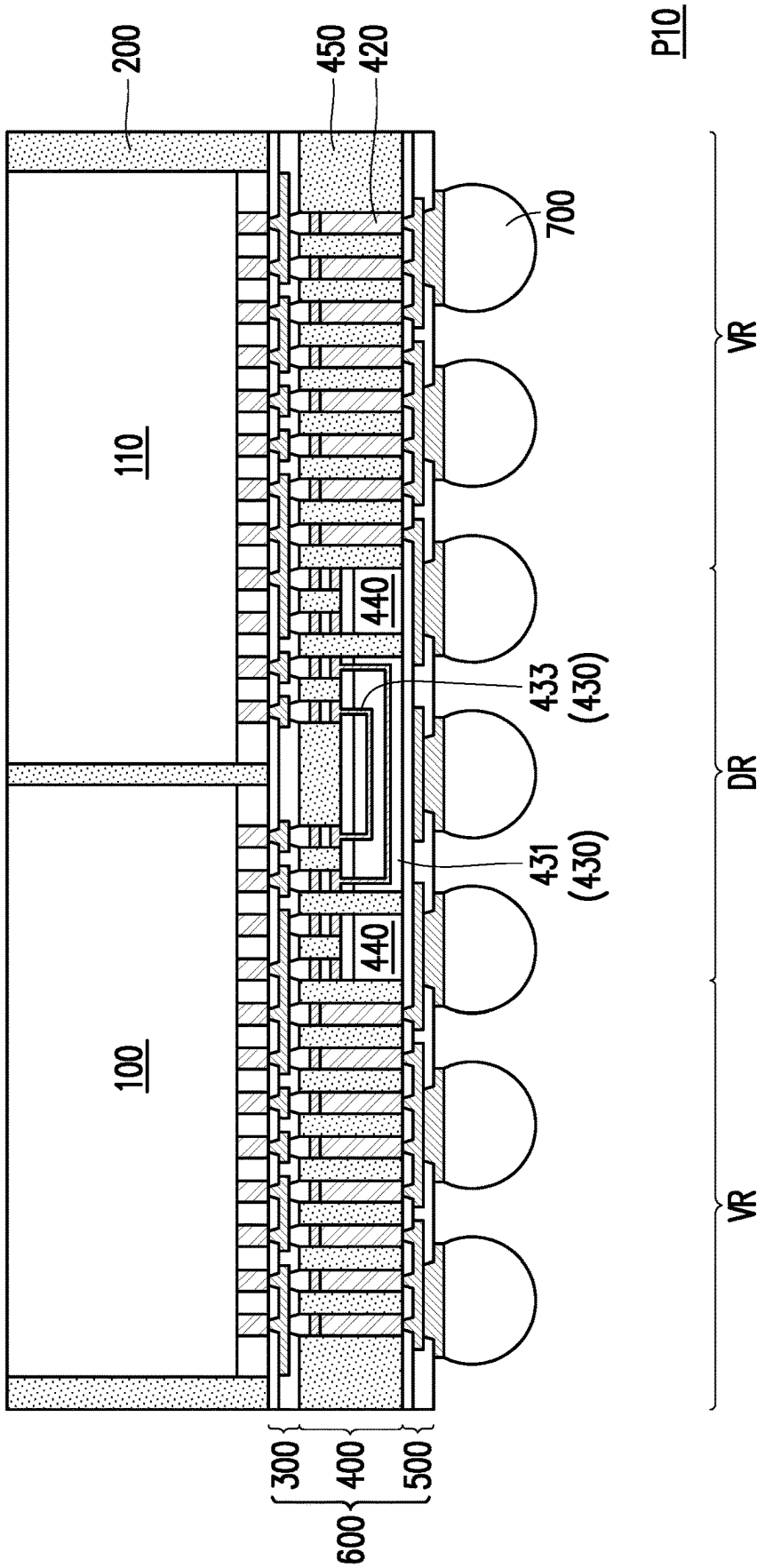

200
450
420
P10
700
110
440
433
(430)
431
(430)
440
100
VR
DR
VR
300
400
500
600

FIG. 1M

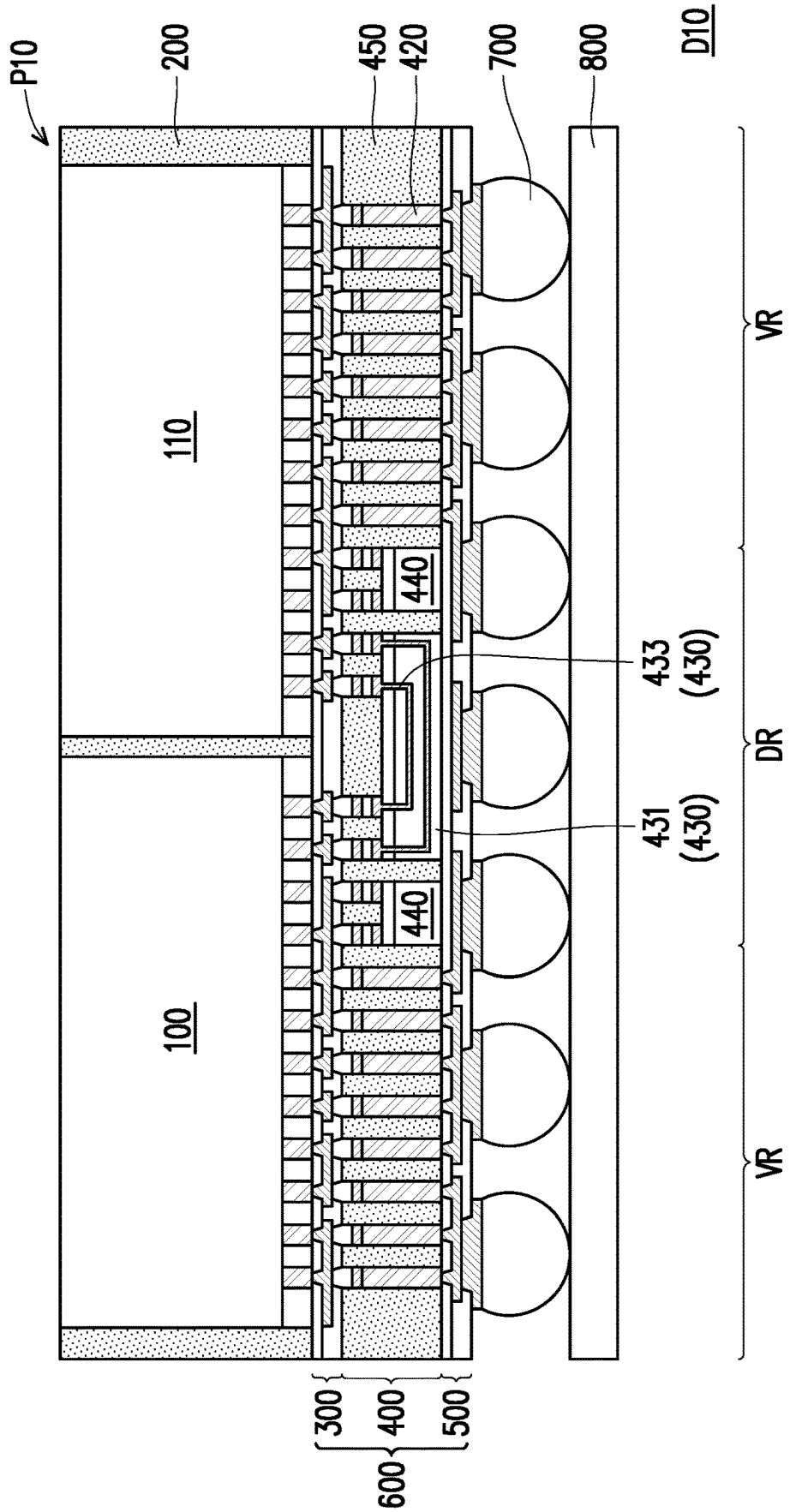

P10
200
450
420
700
800
D10
110
100
440
440
433
(430)
431
(430)
VR
DR
VR
300
400
500
600

FIG. 1N

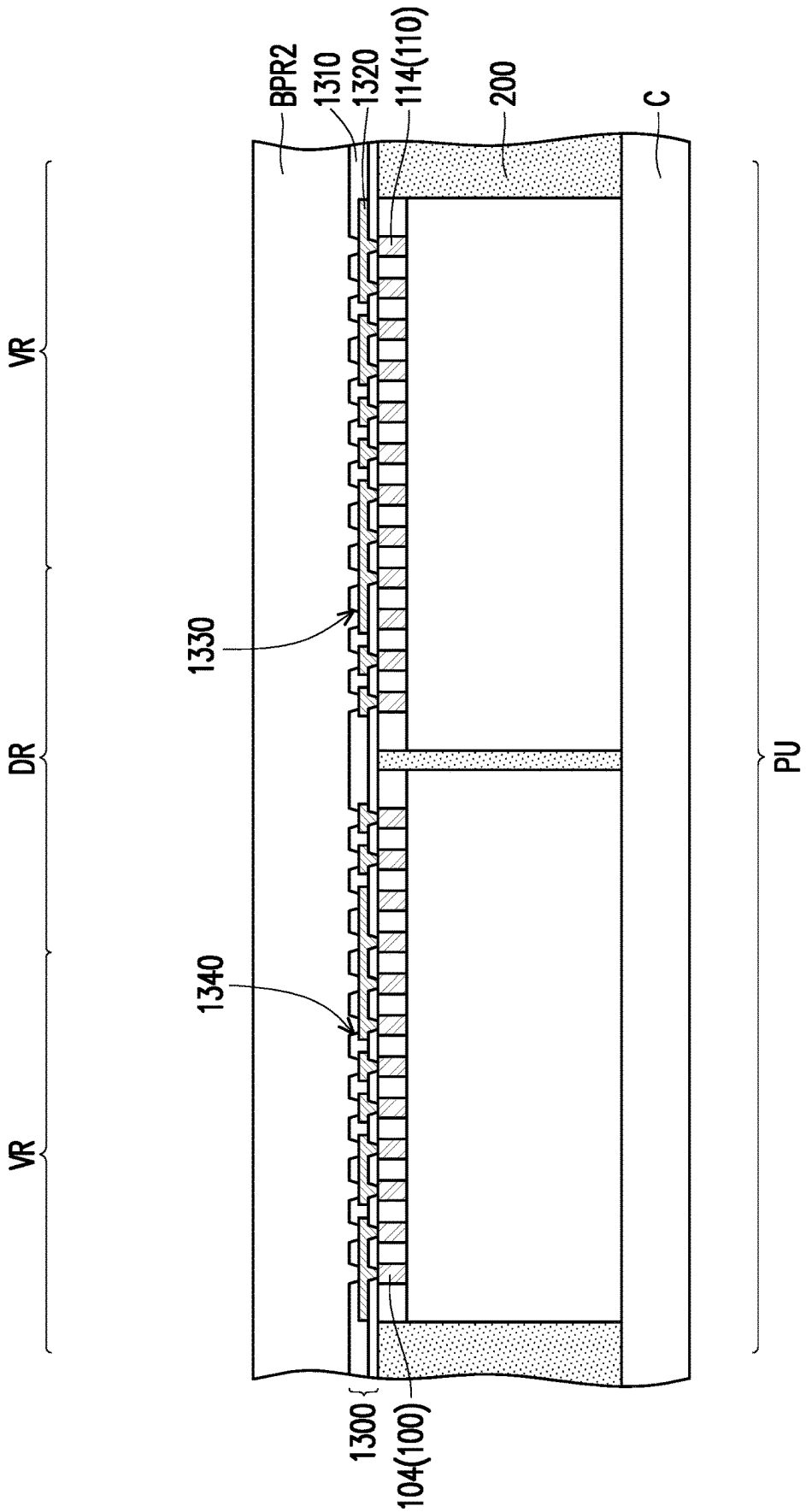


FIG. 2A

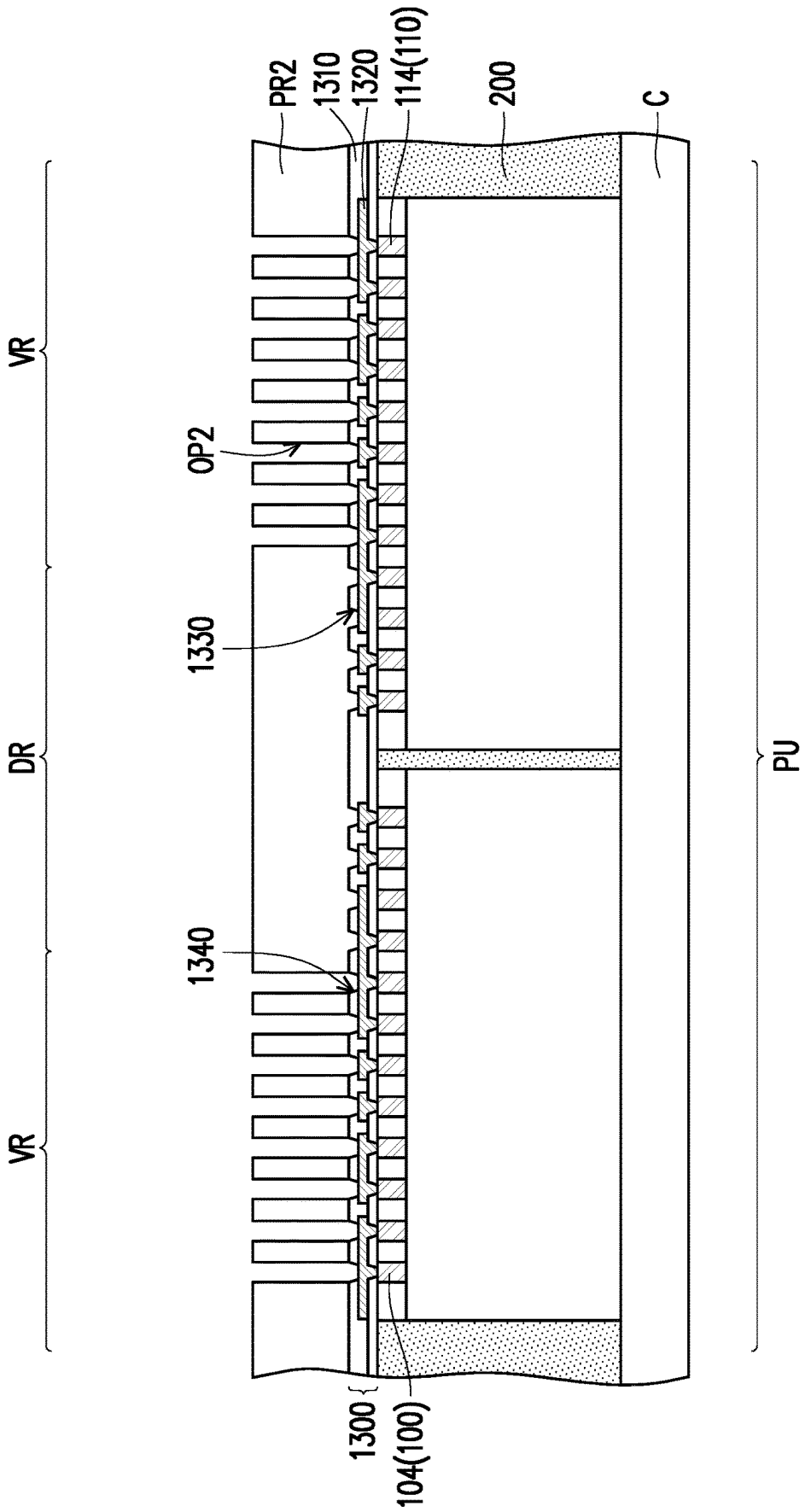
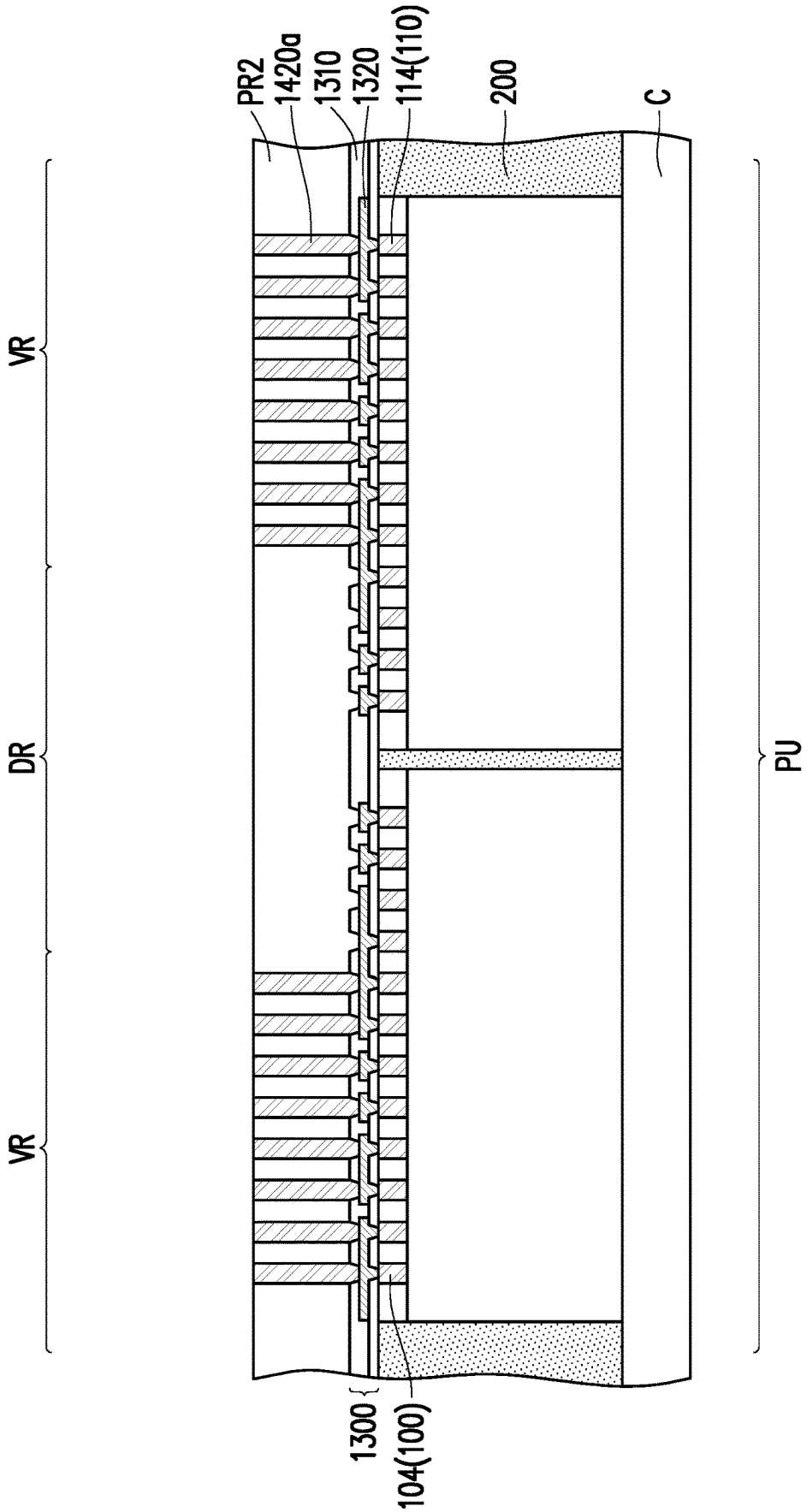

PR2
1310
1320
114(110)
200
C
VR
OP2
1330
DR
1340
VR
PU
1300
104(100)

FIG. 2B



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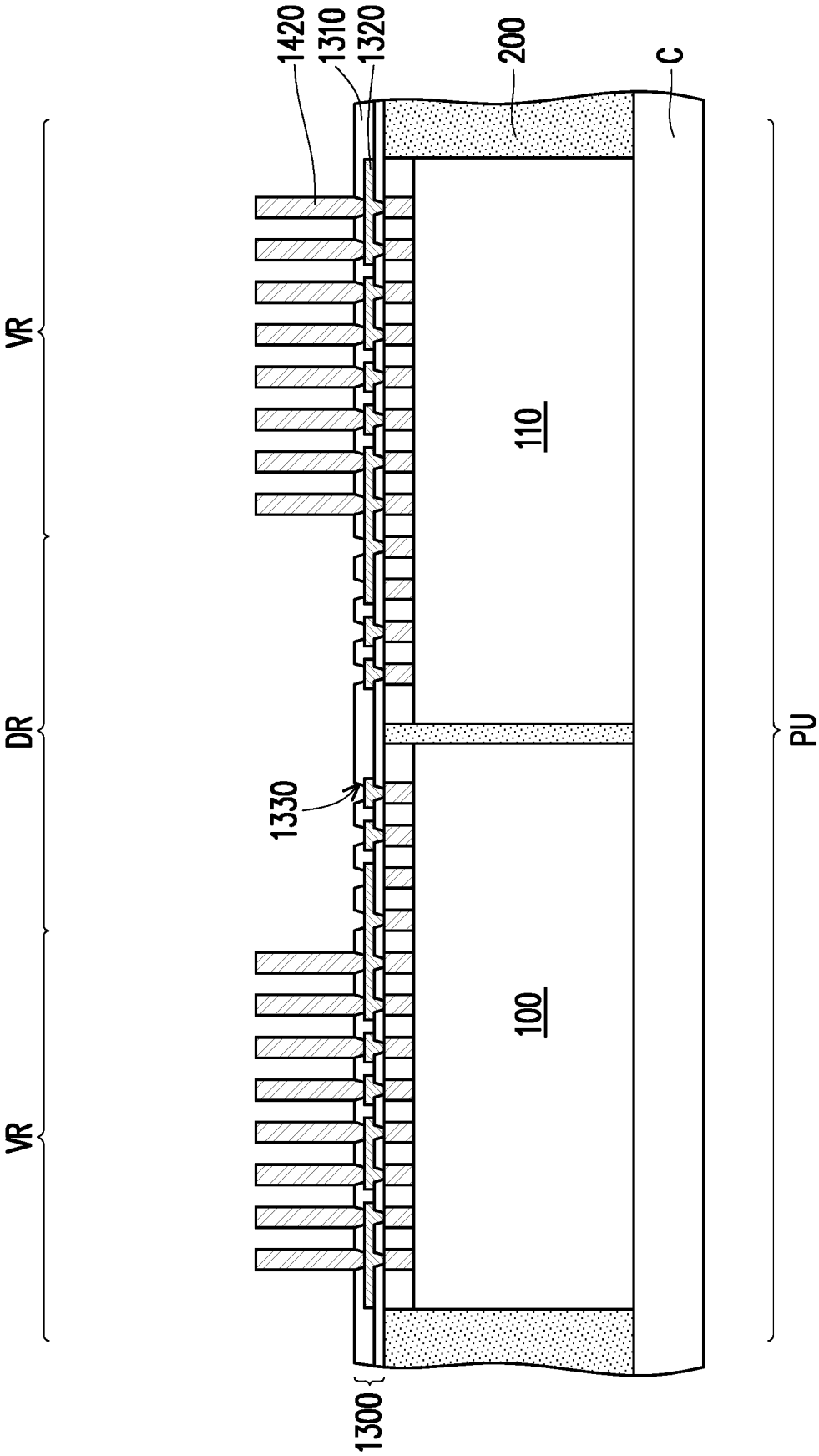

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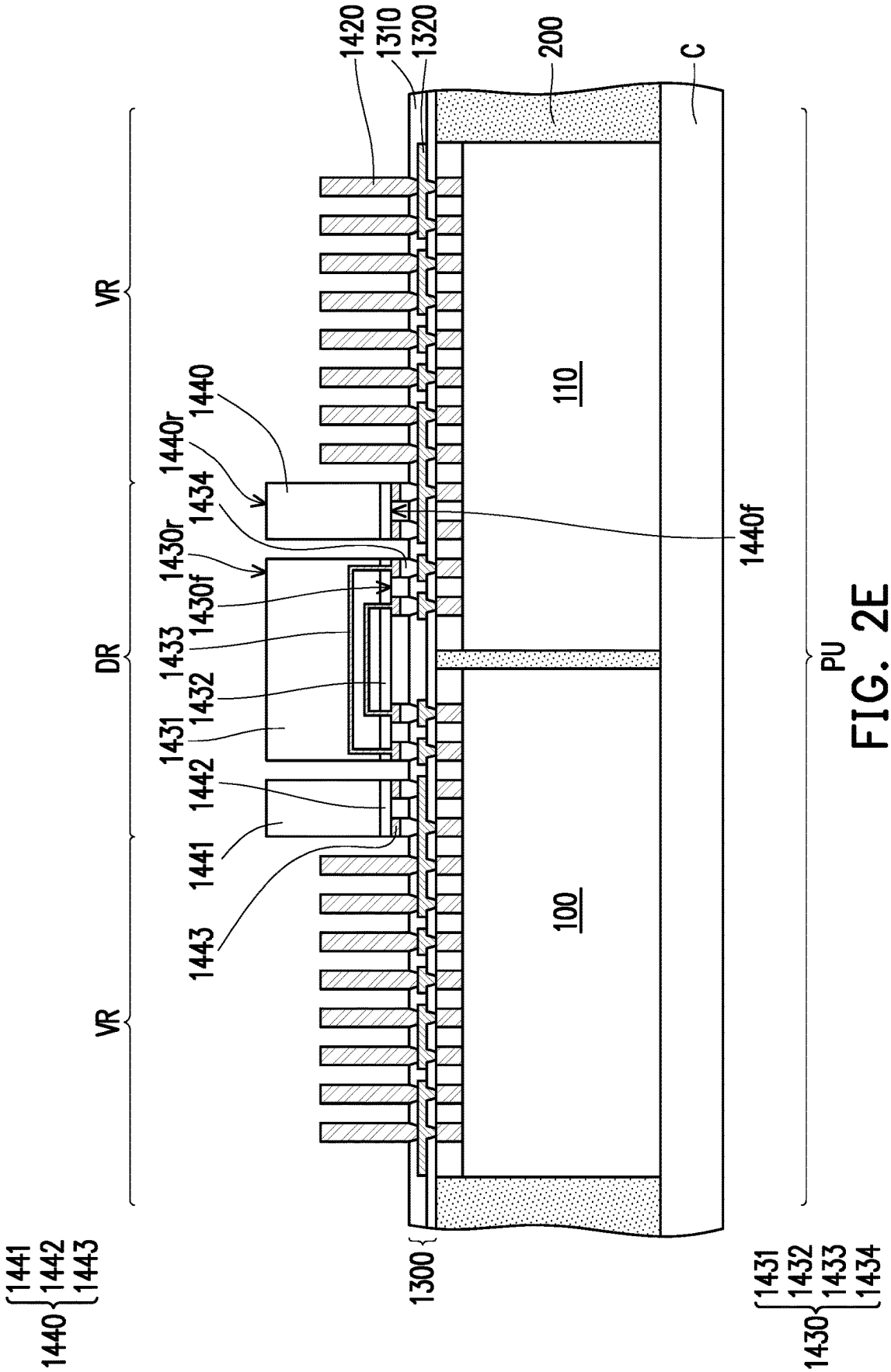

1420
1310
1320
200
C
VR
DR
VR
PU
1440
1440r
1434
1430r
1430f
1433
1432
1431
1442
1441
1443
110
100
1440f
1441
1442
1443
1440
1300
1431
1432
1433
1434
1430

FIG. 2E

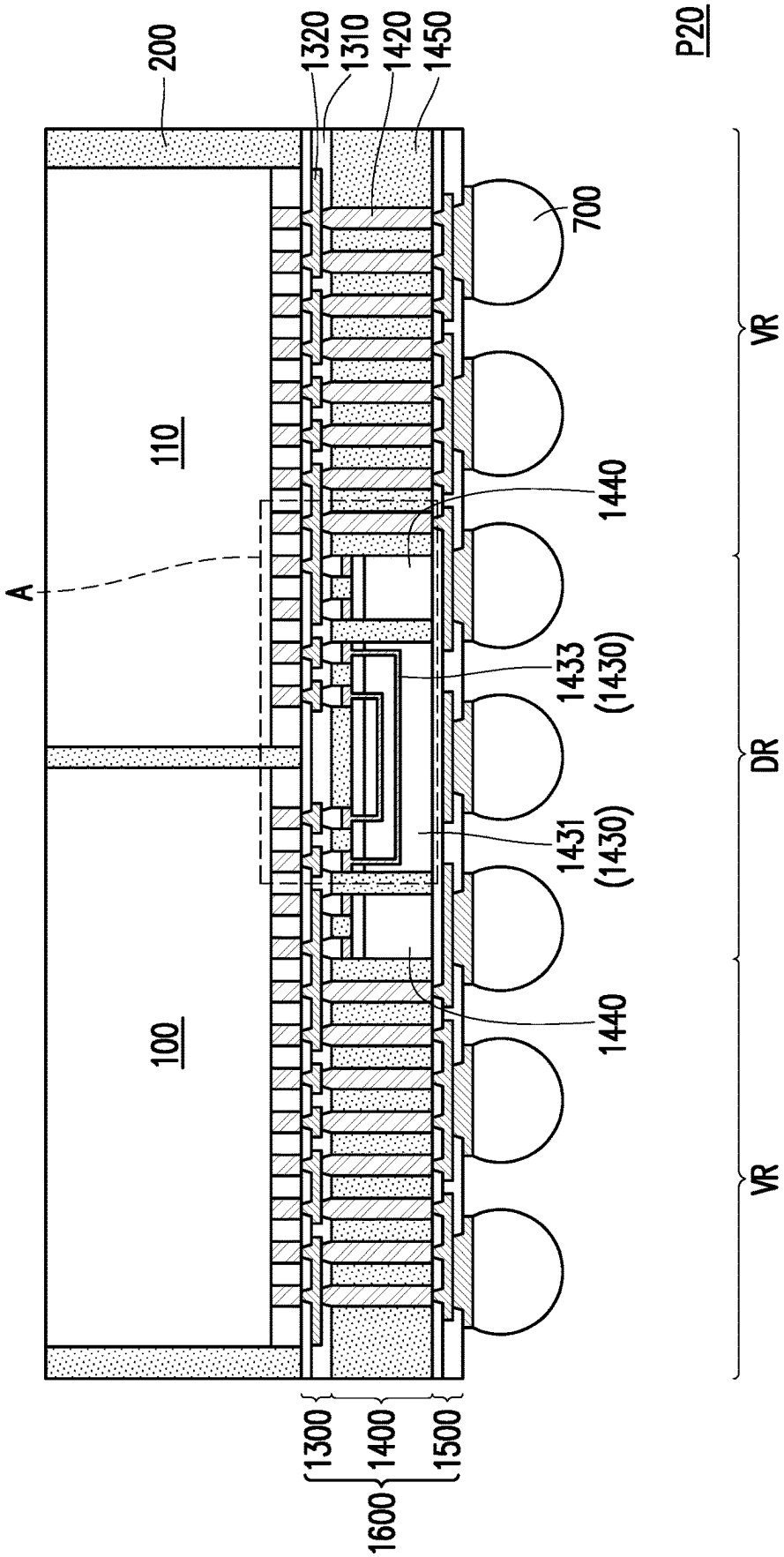

200
1320
1310
1420
1450
P20
700
VR
110
1440
A
1433
(1430)
DR
1431
(1430)
100
1440
VR
1300
1400
1500
1600

FIG. 2F

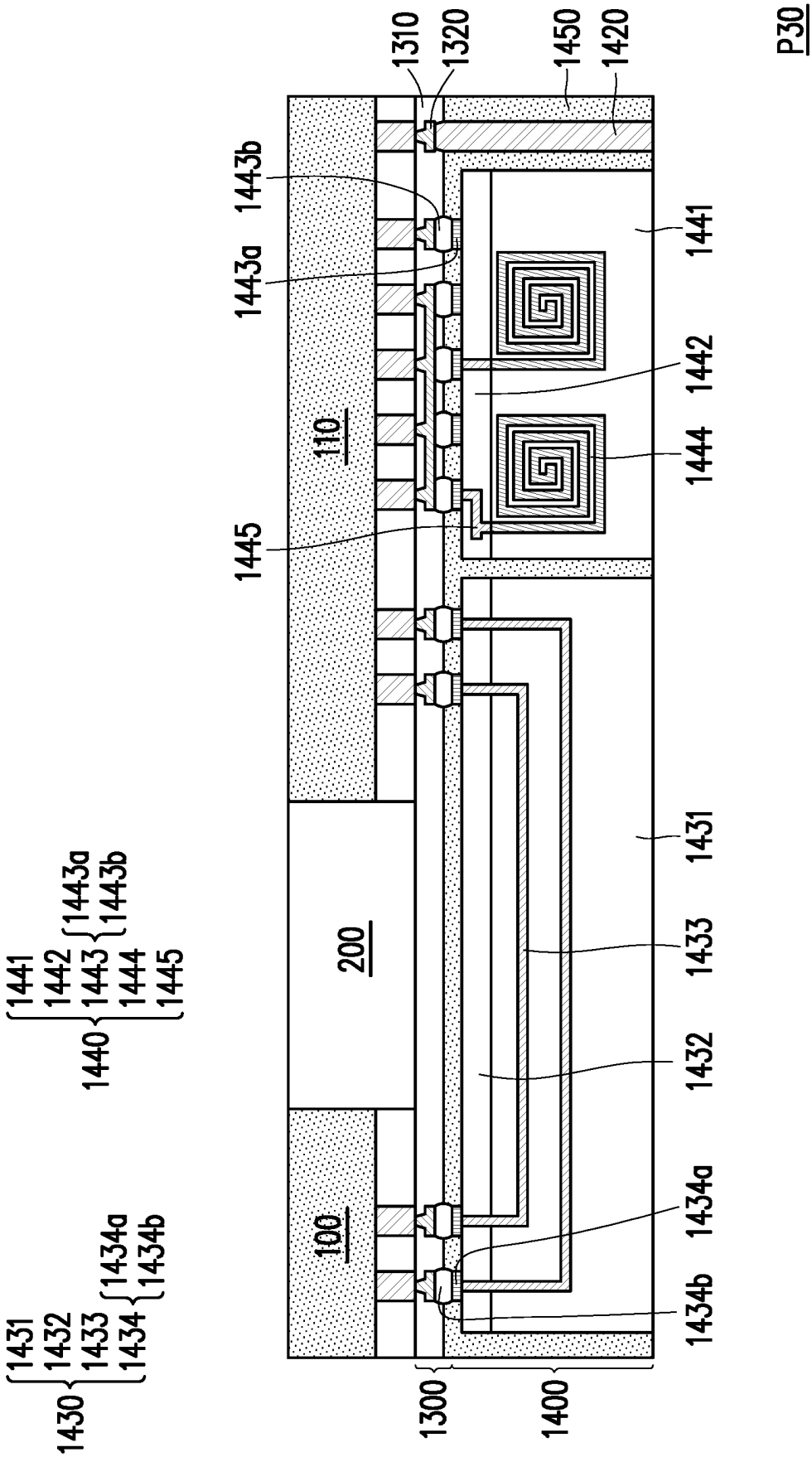
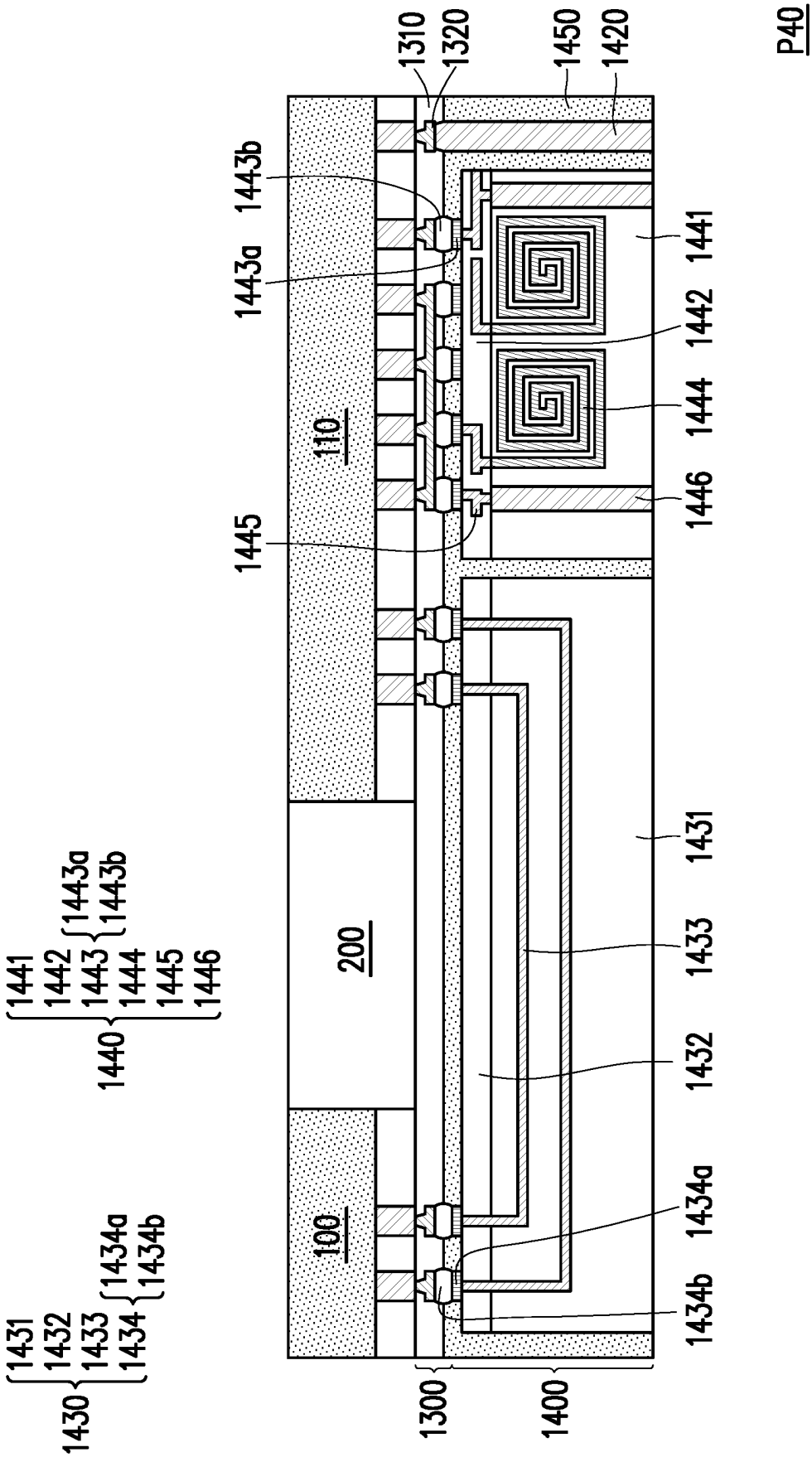

1310
1320
1450
1420
P30
1443b
1443a
1441
1442
1444
110
1445
1431
1433
1432
1434a
1434b
200
100
1441
1442
1443
1444
1445
1440
1443a
1443b
1431
1432
1433
1434
1430
1434a
1434b
1300
1400

FIG. 3A


P40
1310
1320
1450
1420
1443b
1443a
1441
1442
1444
1446
1445
110
200
100
1431
1433
1432
1434a
1434b
1300
1400
1441
1442
1443
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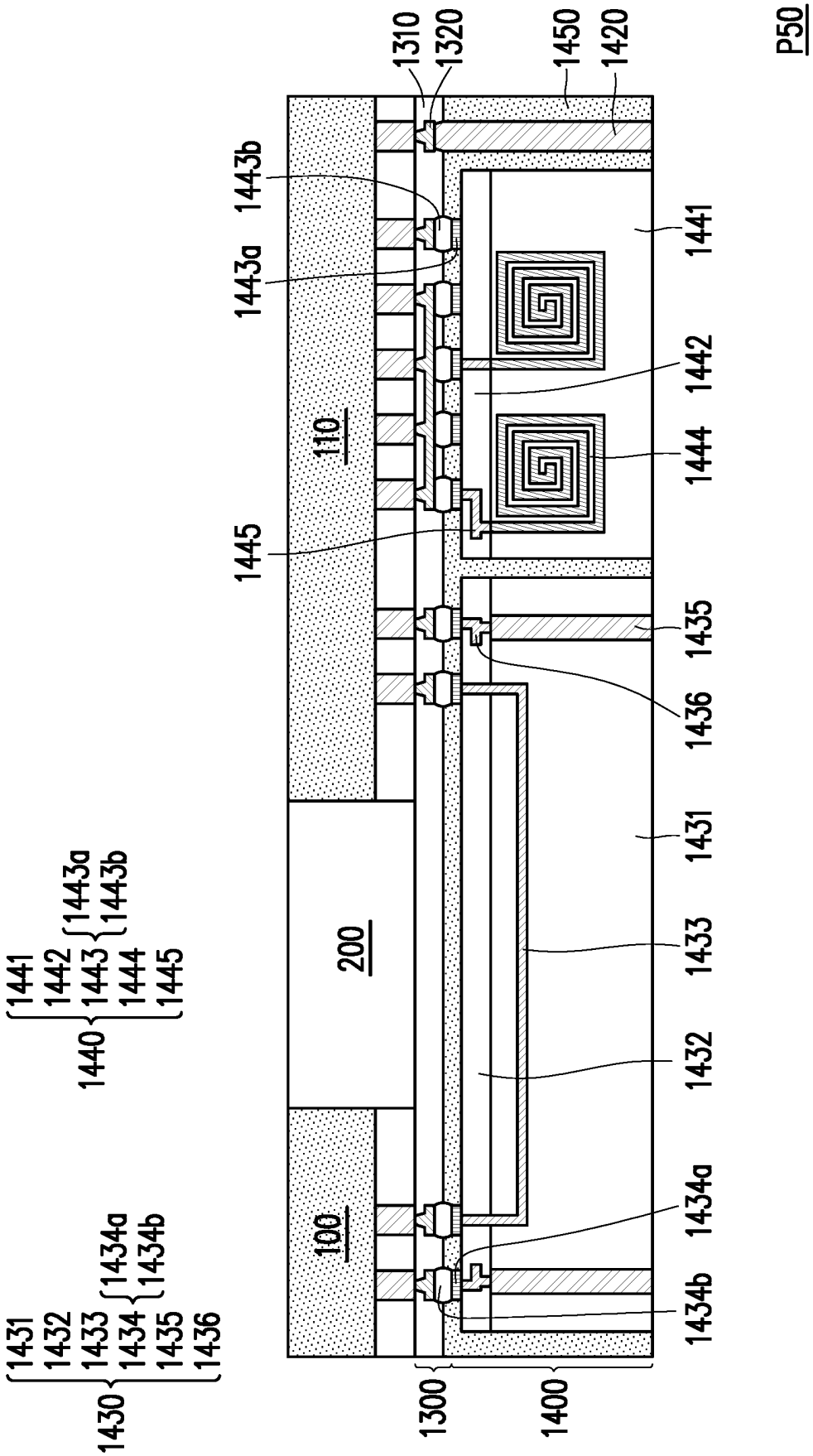
1434a
1434b

FIG. 3B


P50
1310
1320
1450
1420
1443b
1443a
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1442
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1435
1436
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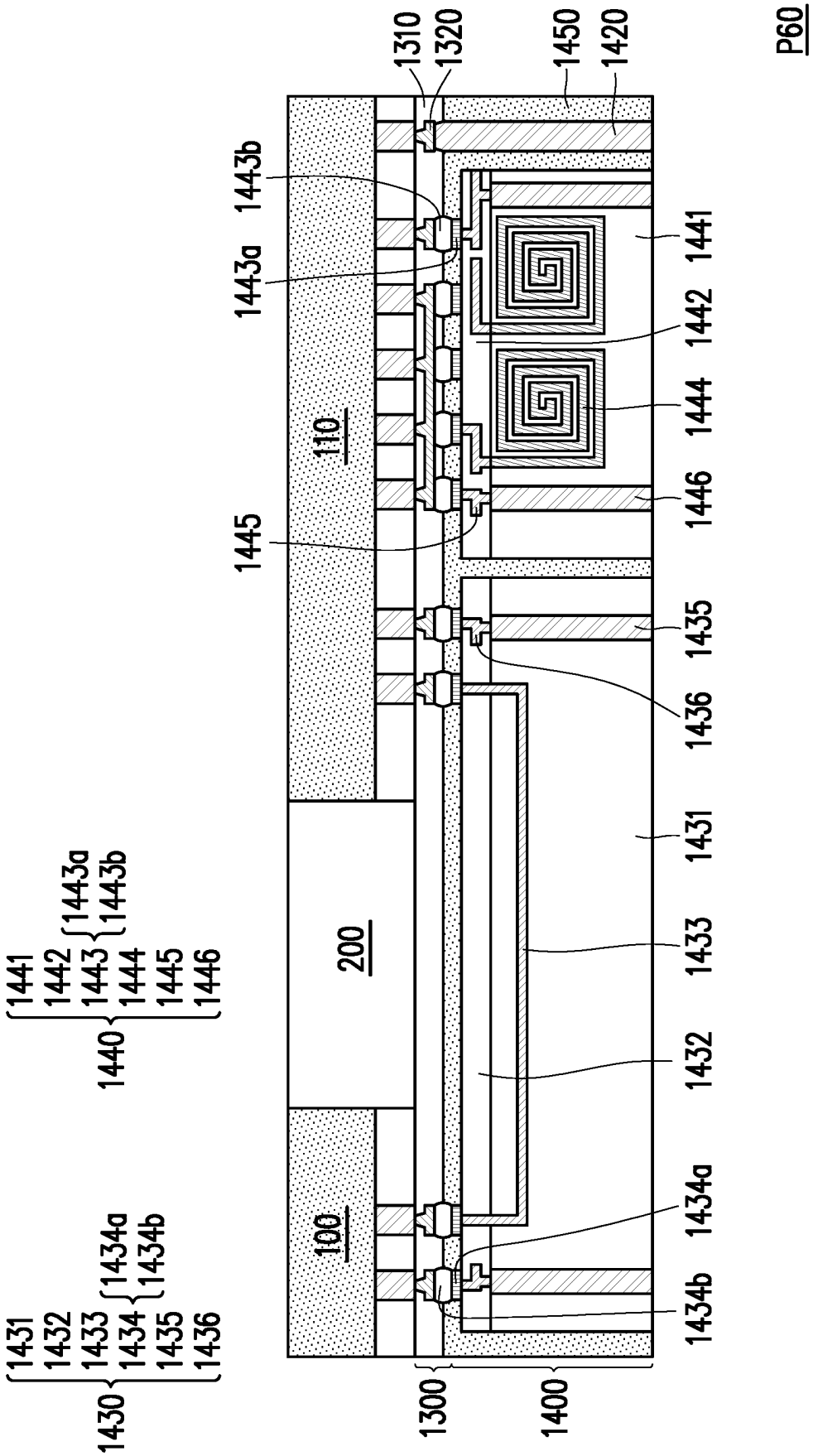
1432
1434a
1434b
110
200
100
1300
1400
1441
1442
1443
1444
1445
1440
1443a
1443b
1431
1432
1433
1434
1435
1436
1430
1434a
1434b

FIG. 3C


P60
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1443b
1443a
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1442
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1445
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1436
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1434a
1434b
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1400
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1444
1445
1446
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1443b
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1435
1436
1430
1434a
1434b

FIG. 3D

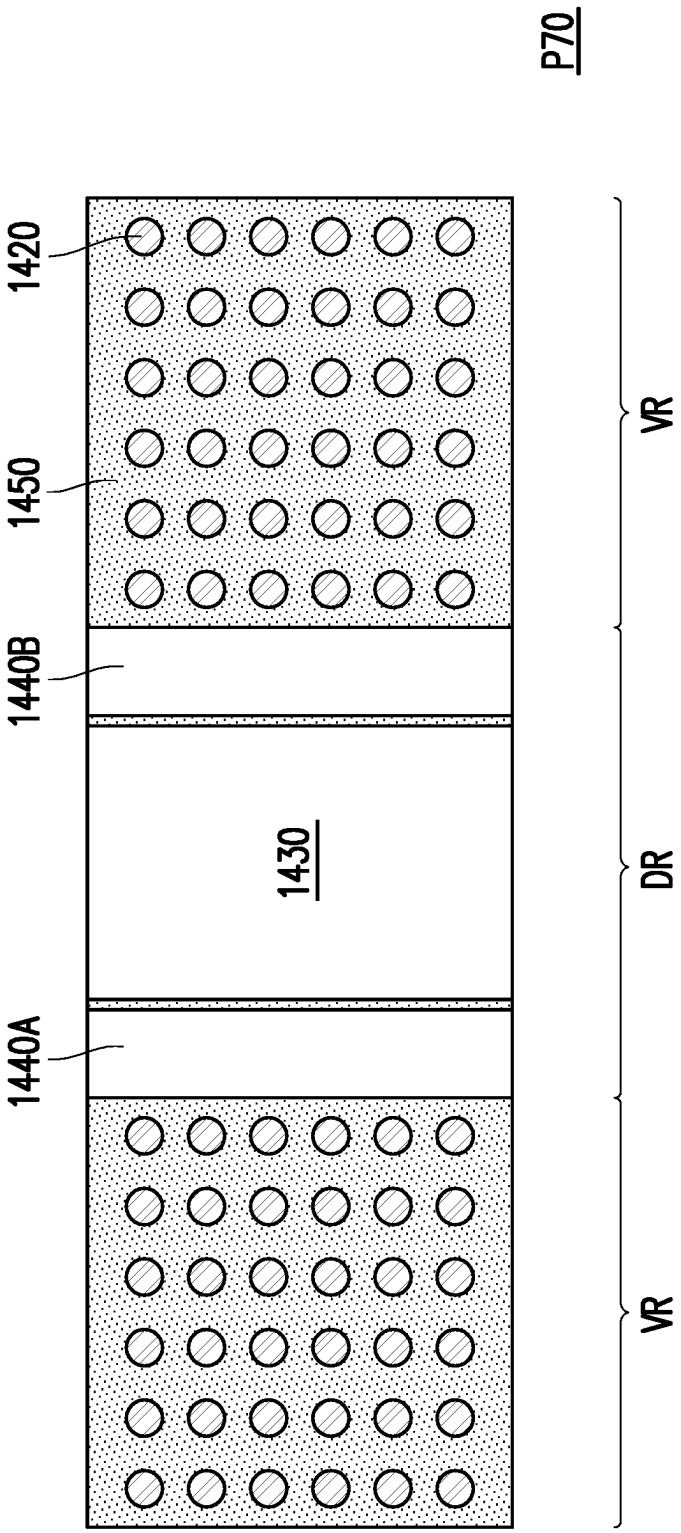


FIG. 4A

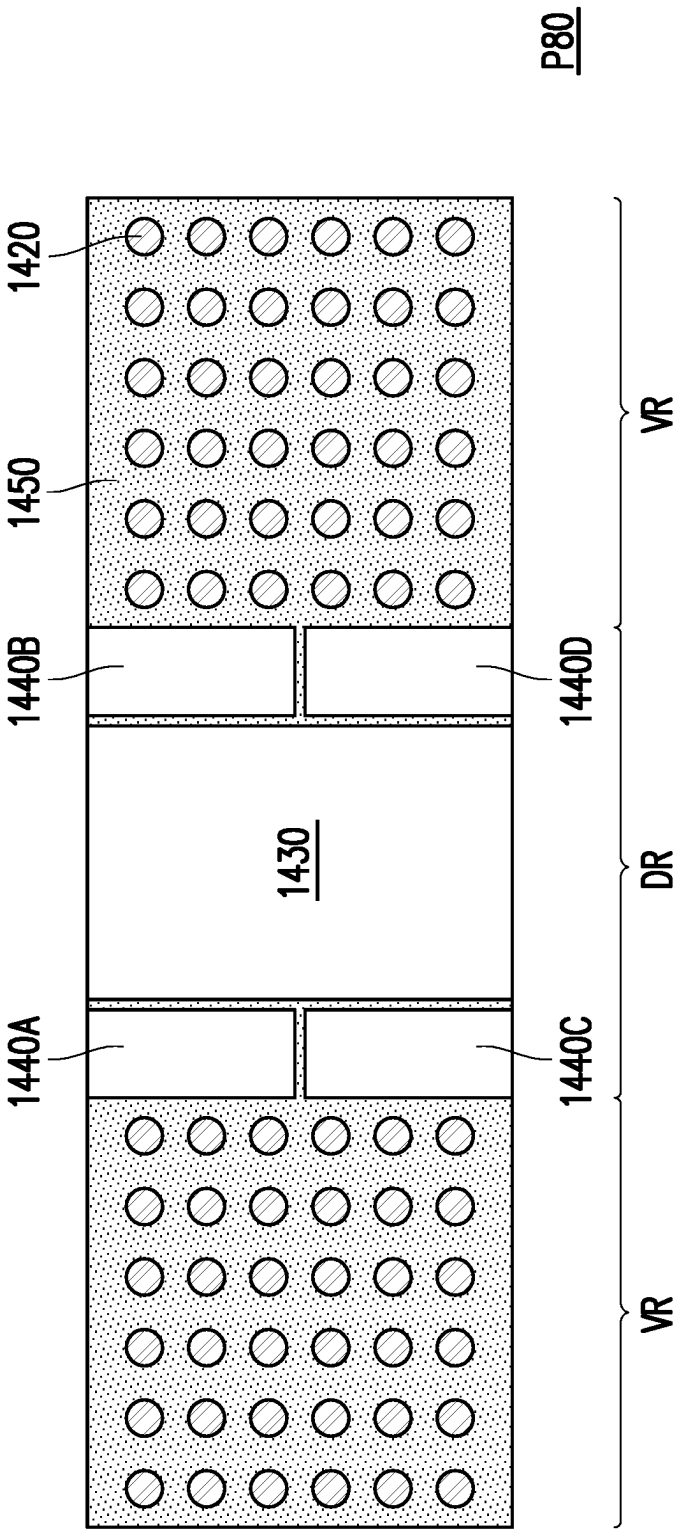


FIG. 4B

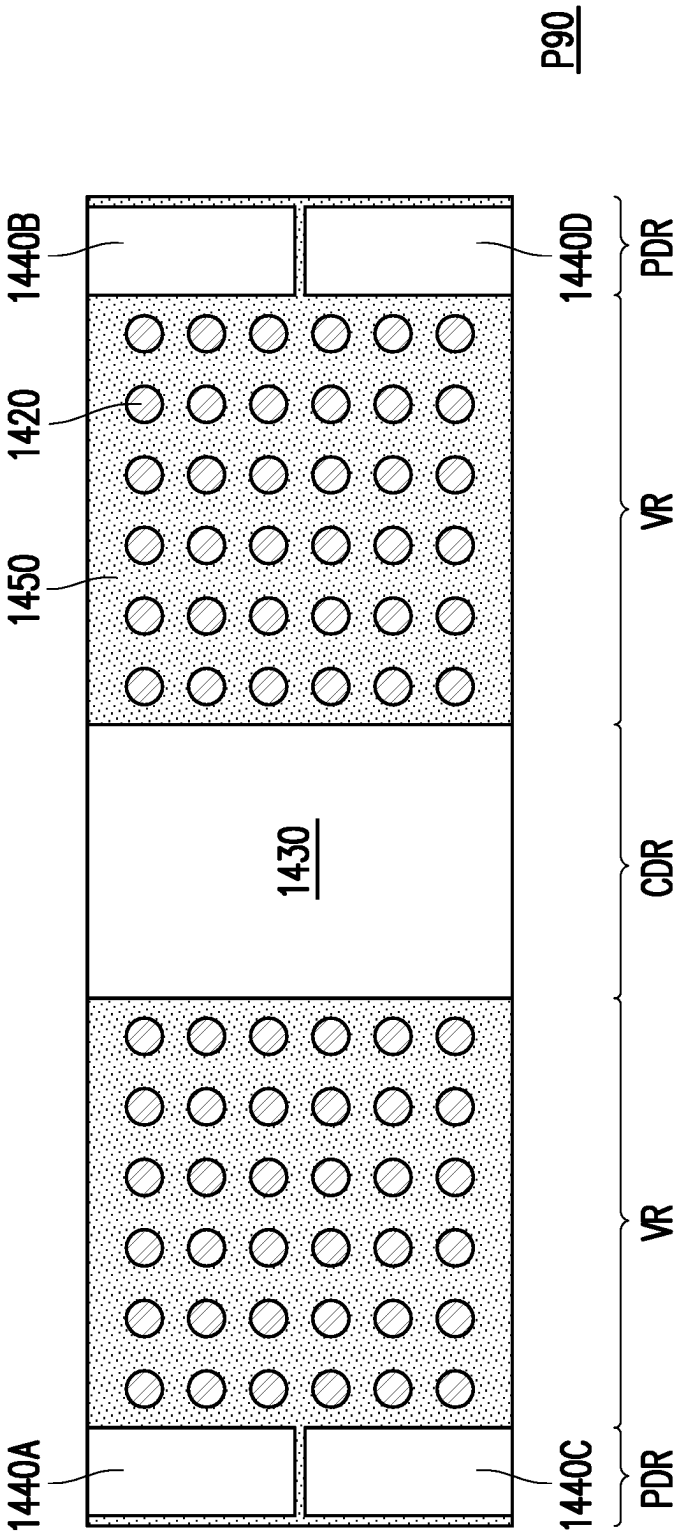
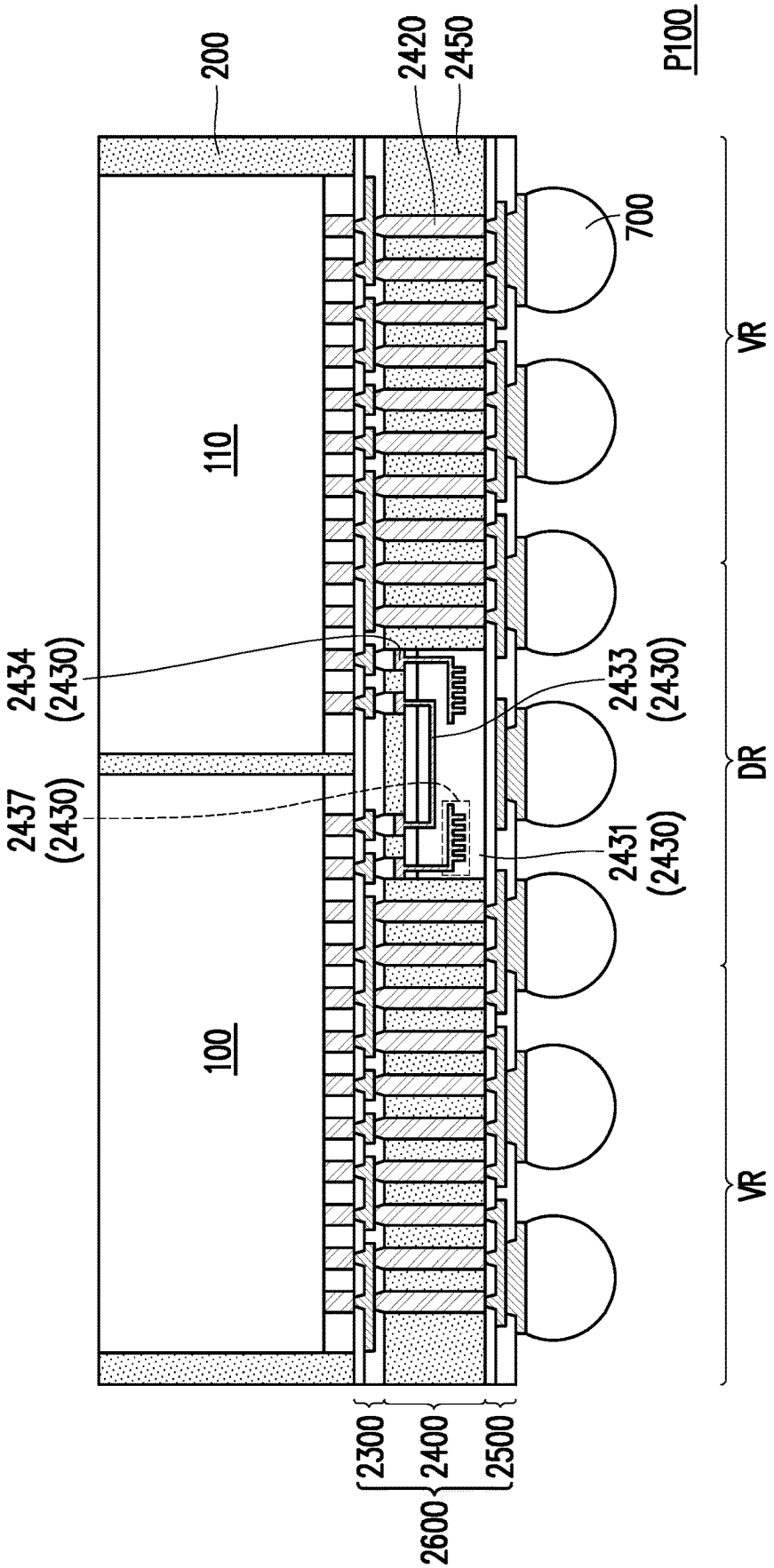


FIG. 4C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2420
2450
P100
700
VR
110
2434
(2430)
2433
(2430)
DR
2437
(2430)
2431
(2430)
100
VR
2300
2400
2500
2600

FIG. 5

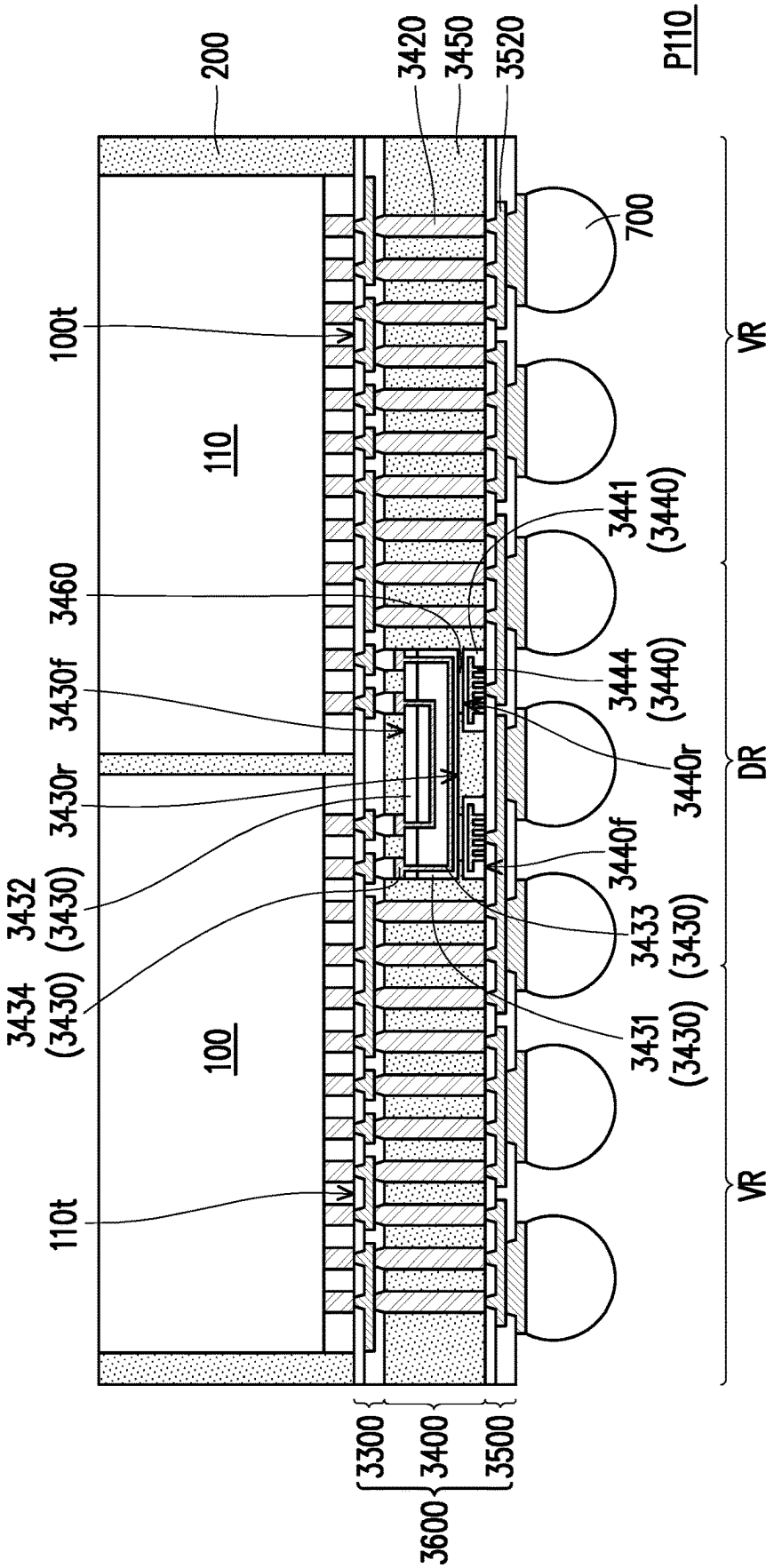
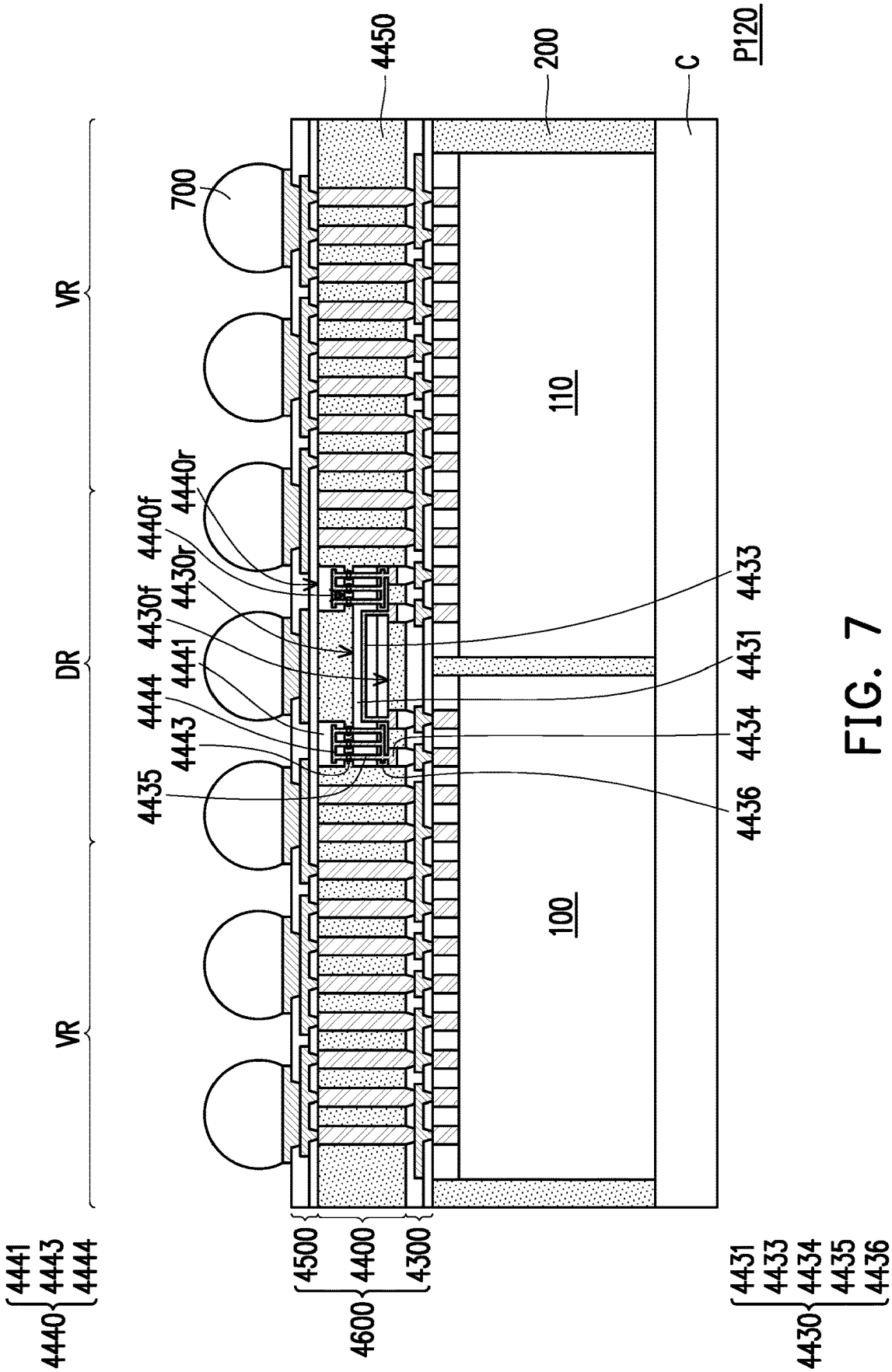

200
3420
3450
3520
P110
700
100t
110
3441
(3440)
3460
3444
(3440)
3430f
3440r
3430r
3440f
3432
(3430)
3433
(3430)
3434
(3430)
3431
(3430)
100
110t
VR
DR
VR
3300
3400
3500
3600

FIG. 6


4450
200
C
P120
700
VR
DR
VR
110
100
4440r
4440f
4430r
4430f
4441
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4443
4435
4433
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4434
4436
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FIG. 7

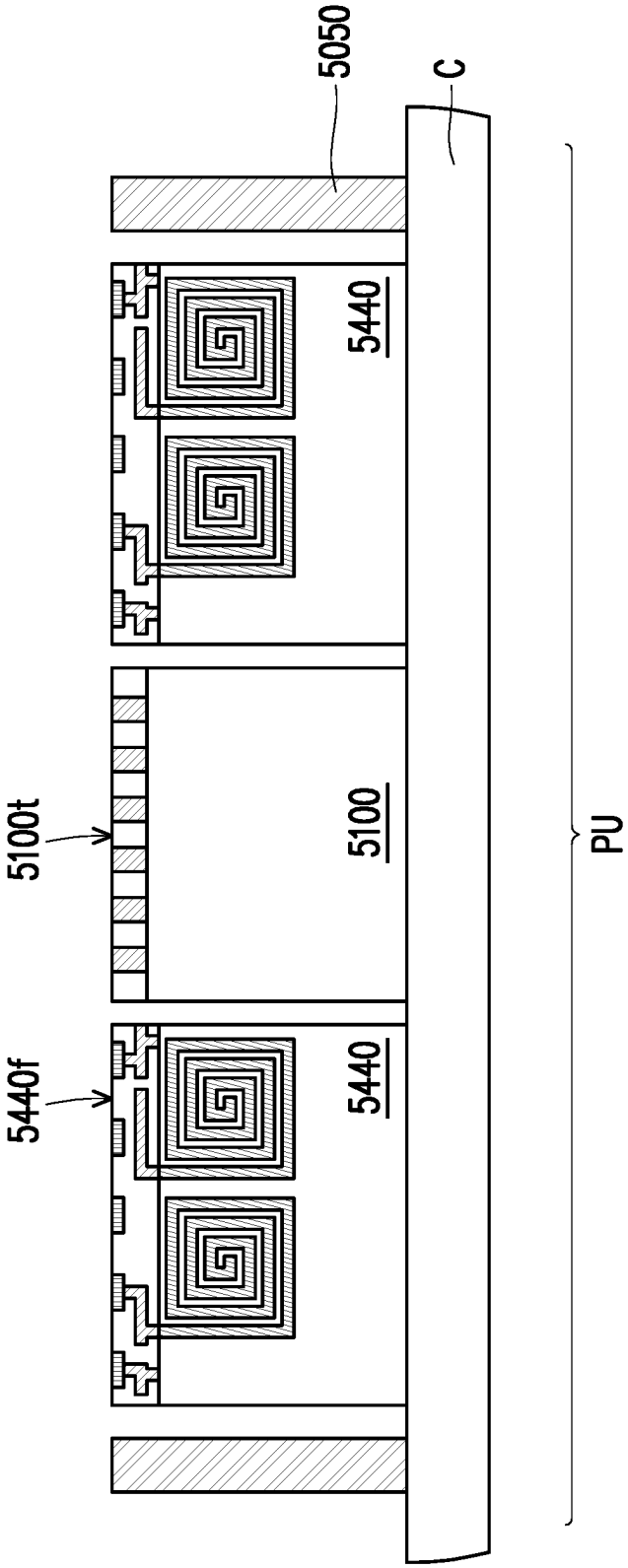

FIG. 8A

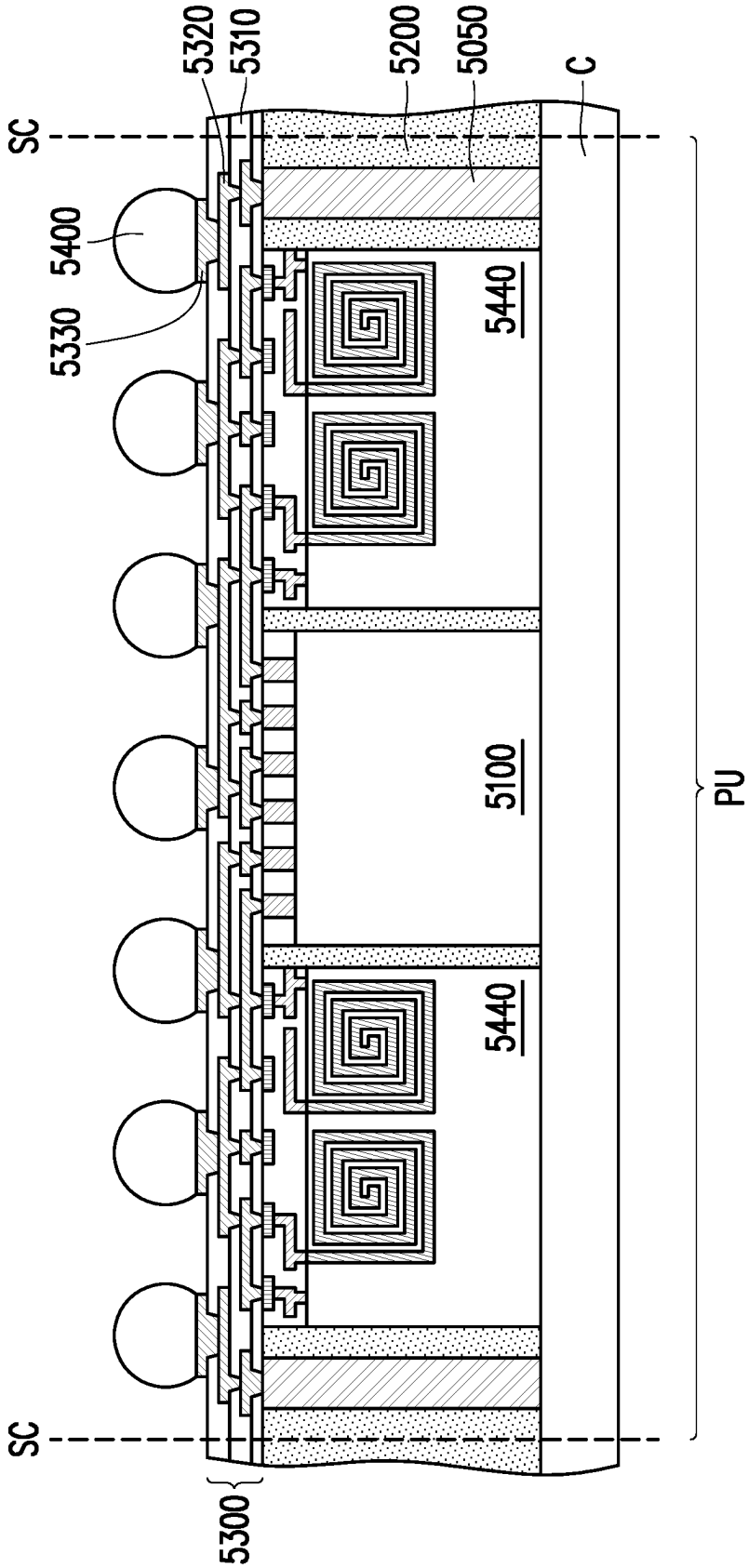

FIG. 8B

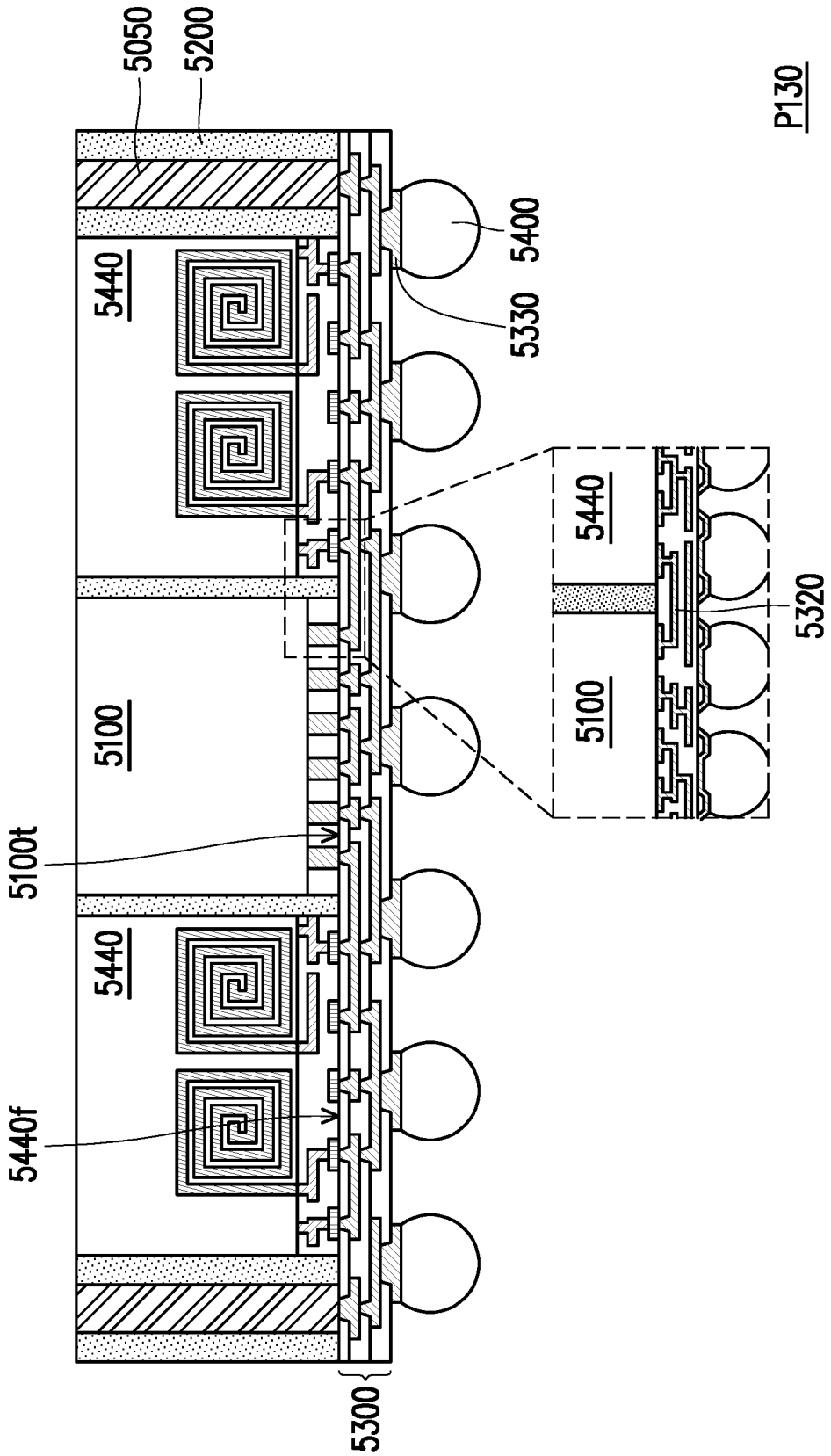
5050
5200
5440
5100
5100t
5440
5440f
5300
5400
5330
5440
5100
5320
P130

FIG. 8C

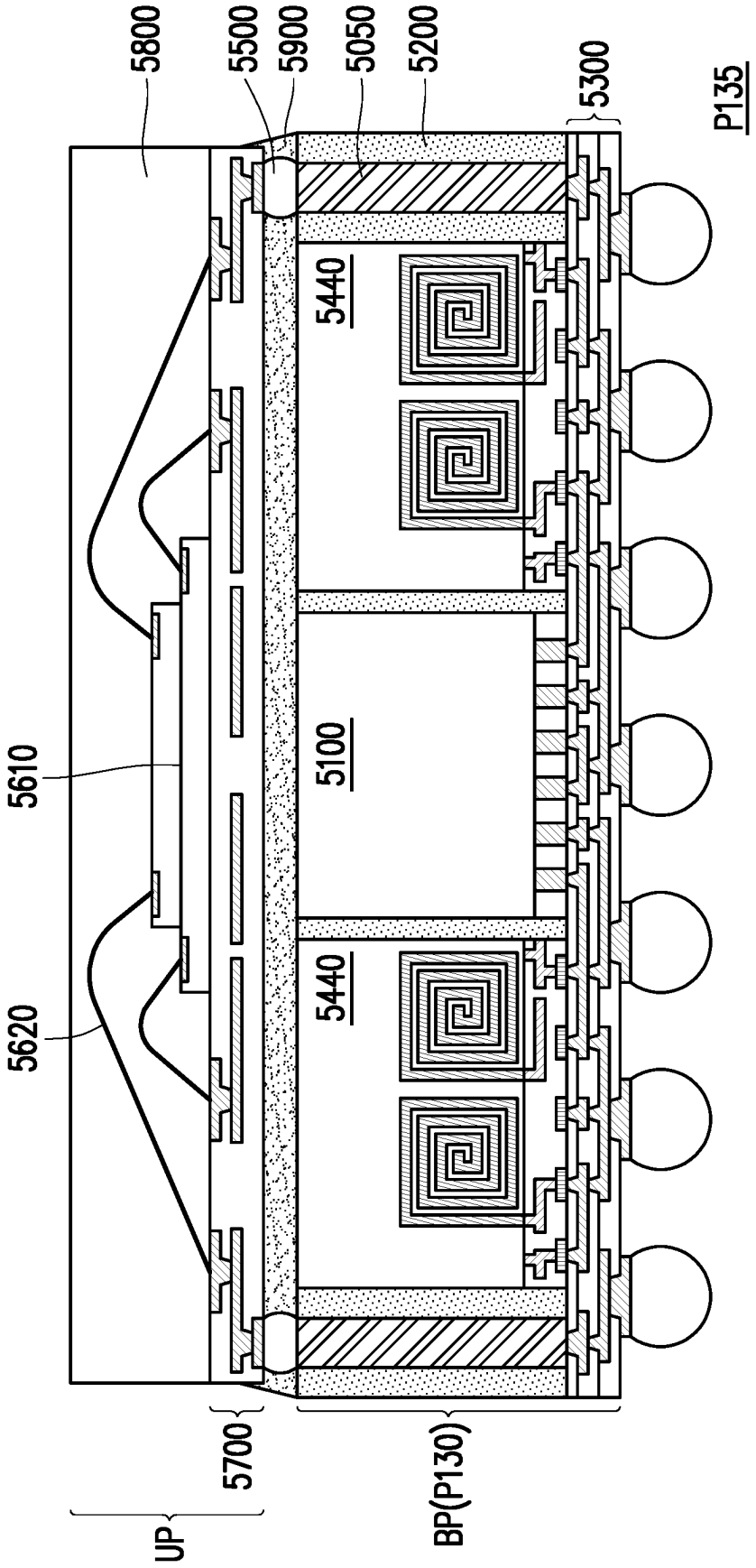
5800
5500
5900
5050
5200
5300
P135
5440
5100
5440
5610
5620
5700
UP
BP(P130)

FIG. 8D

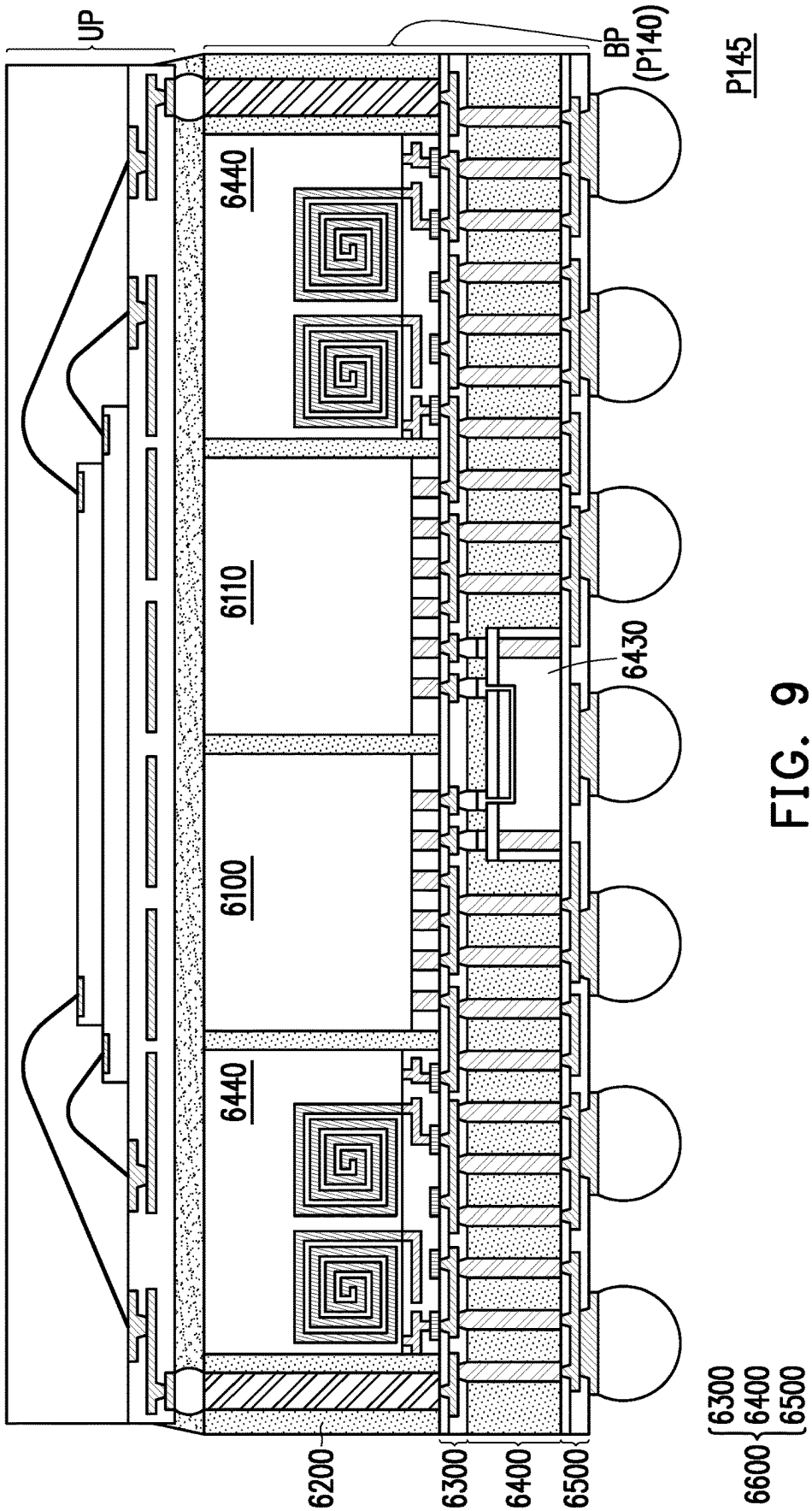
UP
BP
(P140)
P145
6440
6110
6100
6440
6430
6200
6300
6400
6500
6600
6300
6400
6500

FIG. 9

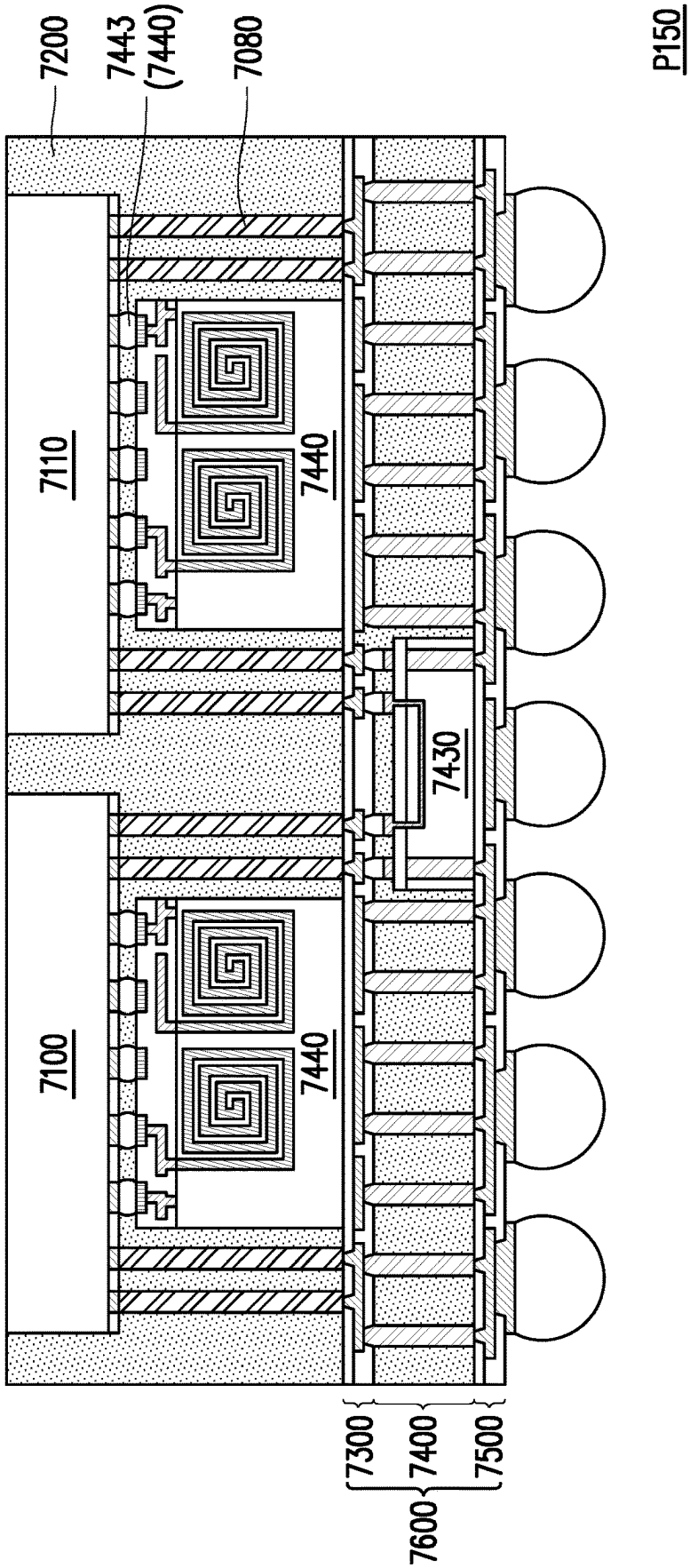
7200
7443
(7440)
7080
P150
7110
7440
7430
7100
7440
7300
7400
7500
7600

FIG. 10

SEMICONDUCTOR PACKAGE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application of and claims the priority of a prior application Ser. No. 16/805,856, filed on Mar. 2, 2020. The prior application Ser. No. 16/805,856 claims the priority benefit of U.S. provisional application Ser. No. 62/905,431, filed on Sep. 25, 2019. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Semiconductor devices and integrated circuits used in a variety of electronic apparatus, such as cell phones and other mobile electronic equipment, are typically manufactured on a single semiconductor wafer. The dies of the wafer may be processed and packaged with other semiconductor devices or dies at the wafer level, and various technologies and applications have been developed for wafer level packaging. Integration of multiple semiconductor devices has become a challenge in the field. To respond to the increasing demand for miniaturization, higher speed, and better electrical performance (e.g., lower transmission loss and insertion loss), more creative packaging and assembling techniques are actively researched.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1**A to FIG. **1**M are schematic cross-sectional views illustrating structures produced during a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.

FIG. **1**N is a schematic cross-sectional view illustrating an electronic device in accordance with some embodiments of the disclosure.

FIG. **2**A to FIG. **2**F are schematic cross-sectional views illustrating structures produced during a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.

FIG. **3**A to FIG. **3**D are schematic cross-sectional views illustrating enlarged portions of semiconductor packages in accordance with some embodiments of the disclosure.

FIG. **4**A to FIG. **4**C are schematic cross-sectional views of semiconductor packages in accordance with some embodiments of the disclosure.

FIG. **5** through FIG. **7** are schematic cross-sectional views illustrating semiconductor packages in accordance with some embodiments of the disclosure.

FIG. **8**A through FIG. **8**C are schematic cross-sectional views illustrating structures produced during a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.

FIG. **8**D is a schematic cross-sectional view illustrating a semiconductor package according to some embodiments of the disclosure.

FIG. **9** is a schematic cross-sectional view illustrating a semiconductor package in accordance with some embodiments of the disclosure.

FIG. **10** is a schematic cross-sectional view illustrating a semiconductor package in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

FIG. **1**A to FIG. **1**M are schematic cross-sectional views illustrating structures produced during a manufacturing process of a semiconductor package SP**10** in accordance with some embodiments of the disclosure. Referring to FIG. **1**A, a carrier C is provided. In some embodiments, the carrier C is a glass substrate, a metal plate, a plastic supporting board or the like, but other suitable substrate materials may be used as long as the materials are able to withstand the subsequent steps of the process. In some embodiments, a de-bonding layer (not shown) may be formed over the carrier C. In some embodiments, the de-bonding layer includes a light-to-heat conversion (LTHC) release layer, which facilitates peeling the carrier C away from the semiconductor package when required by the manufacturing process.

In some embodiments, semiconductor dies **100**, **110** are disposed side by side over the temporary carrier C with a pick-and-place process. In some embodiments, the semiconductor dies **100** include a semiconductor substrate **102**, one

or more contact pads **104**, and a passivation layer **106**. The contact pads **104** may be formed on a top surface **102***t* of the semiconductor substrate **102**. The passivation layer **106** may cover the top surface **102***t* and have a plurality of openings that exposes at least a portion of each contact pad **104**. In some embodiments, the semiconductor dies **100** may further include conductive posts (not shown) electrically connected to the contact pads **104** and a protective layer (not shown) disposed on the passivation layer **106** and surrounding the conductive posts. In some embodiments the passivation layer **106** may completely cover the contact pads **104** and (temporarily) constitute a top surface **100***t* of the semiconductor die **100**. The semiconductor dies **110** may include similar components as just described for the semiconductor dies **100**. For example, each semiconductor die **110** may include a semiconductor substrate **112**, contact pads **114** and a passivation layer **116**.

In some embodiments, the semiconductor dies **100**, **110** are placed on the carrier C with the top surfaces **100***t*, **110***t* where the contact pads **104**, **114** are located facing away from the carrier C. Backside surfaces **100***b*, **110***b* of the semiconductor dies **100**, **110** opposite to the top surfaces **100***t*, **110***t* may be in contact with the carrier C (or the die attach film, if included). In some embodiments, the semiconductor dies **100**, **110** included in a semiconductor package may have different sizes, or components of different sizes. For example, the semiconductor dies **100**, **110** may differ for the thickness of the semiconductor substrates **102**, **112**, the number of contact pads **104**, **114**, whether conductive posts are included or not, and so on.

In some embodiments, the semiconductor substrates **102**, **112** are made of semiconductor materials, such as semiconductor materials of the groups III-V of the periodic table. In some embodiments, the semiconductor substrates **102**, **112** include elementary semiconductor materials such as silicon or germanium, compound semiconductor materials such as silicon carbide, gallium arsenide, indium arsenide, or indium phosphide or alloy semiconductor materials such as silicon germanium, silicon germanium carbide, gallium arsenide phosphide, or gallium indium phosphide. In some embodiments, the semiconductor substrates **102**, **112** include active components (e.g., transistors or the like) and optionally passive components (e.g., resistors, capacitors, inductors, or the like) formed therein. In certain embodiments, the contact pads **104**, **114** include aluminum pads, copper pads, or other suitable metal pads. In some embodiments, the passivation layers **106**, **116** may be a single layer or a multi-layered structure, including a silicon oxide layer, a silicon nitride layer, a silicon oxy-nitride layer, a dielectric layer formed by other suitable dielectric materials or combinations thereof.

Each semiconductor die **100**, **110** may independently be or include a logic die, such as a central processing unit (CPU) die, a graphic processing unit (GPU) die, a micro control unit (MCU) die, an input-output (I/O) die, a baseband (BB) die, or an application processor (AP) die. In some embodiments, some of the semiconductor dies **100**, **110** may be memory dies. That is, the semiconductor dies **100**, **110** are or function as active devices.

Referring to FIG. **1**B, an encapsulant **200** is formed over the carrier C to encapsulate the semiconductor dies **100**, **110**. The encapsulant **200** laterally encircles the semiconductor dies **100**, **110**, extending also in the gaps in between the semiconductor dies **100**, **110**. In some embodiments, a material of the encapsulant **200** includes a molding compound, a polymeric material, such as polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzooxazole (PBO), a combination thereof, or other suitable polymer-based dielectric materials. The encapsulant **200** may be formed by a sequence of over-molding and planarization steps. For example, the encapsulant **200** may be originally formed by a molding process (such as a compression molding process) or a spin-coating process to completely cover the semiconductor dies **100**, **110**. In some embodiments, the planarization of the encapsulant **200** includes performing a mechanical grinding process and/or a chemical mechanical polishing (CMP) process. In some embodiments, the planarization process is performed until the contact pads **104**, **114** of the semiconductor dies **100**, **110** are exposed. In some embodiments, portions of the passivation layers **106**, **116** are also removed during the planarization process of the encapsulant **200**. In some embodiments, following the planarization process, the top surfaces **100***t*, **110***t* of the semiconductor dies **100**, **110** (the contact surfaces exposing the contact pads **104**, **114** or the conductive posts if included) and the top surface **200***a* of the encapsulant **200** may be substantially at a same level height (be substantially coplanar). With the formation of the encapsulant **200**, a reconstructed wafer RW is obtained. In some embodiments, the reconstructed wafer RW includes a plurality of package units PU. In other words, the exemplary process may be performed at a reconstructed wafer level, so that multiple package units PU are processed in the form of the reconstructed wafer RW. In the cross-sectional view of FIG. **1**B, a single package unit PU is shown for simplicity but, of course, this is for illustrative purposes only, and the disclosure is not limited by the number of package units PU being produced in the reconstructed wafer RW.

Referring to FIG. **1**C, in some embodiments, an inner redistribution layer **300** is formed on the semiconductor dies **100**, **110** and the encapsulant **200** over the carrier C. In some embodiments, the inner redistribution layer **300** includes dielectric layers **310** alternately stacked with one or more metallization tiers **320**. In some embodiments, the dielectric layers **310** are constituted by at least two dielectric layers. The metallization tier **320** includes routing conductive traces sandwiched between pairs of adjacent dielectric layers **310**. In some embodiments, a material of the dielectric layers **310** includes polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzooxazole (PBO), combinations thereof, or any other suitable polymer-based dielectric material. The dielectric layers **310** may be formed by suitable fabrication techniques such as spin-on coating, chemical vapor deposition (CVD), or the like. In some embodiments, a material of the metallization tier **320** includes copper, aluminum, or the like. In some embodiments, the material of the metallization tier **120** includes copper. Throughout the description, the term "copper" is intended to include substantially pure elemental copper, copper containing unavoidable impurities, and copper alloys containing elements such as tantalum, indium, tin, zinc, manganese, chromium, titanium, germanium, strontium, platinum, magnesium, aluminum, or zirconium. The metallization tier **120** may be formed by, for example, electroplating, deposition, and/or photolithography and etching. In some alternative embodiments, more metallization tiers **320** and more dielectric layers **310** than the ones illustrated in FIG. **1**C may be formed depending on production requirements. In these embodiments, each metallization tier is sandwiched between a pair of consecutive dielectric layers.

In some embodiments, the innermost dielectric layer **310** (the dielectric layer **310** closer to the semiconductor dies **100**, **110**) is patterned to include openings exposing portions of the contact pads **104**, **114**. The metallization tier **320** may fill the openings to establish electrical contact with the

semiconductor dies **100**, **110**. In some embodiments, the outermost dielectric layer **310** (the dielectric layer **310** further away from the semiconductor dies **100**, **110**) is patterned to include openings **330** and **340** exposing portions of the (outermost) metallization tier **320**. In some embodiments, the inner redistribution layer **300** includes one or more device attach regions DR in which the openings **330** are located, and one or more interconnect via regions VR in which the openings **340** are located. In some embodiments, the device attach region DR is located towards a central portion of the package unit PU, extending over portions of adjacent semiconductor dies **100**, **110**, and is surrounded by interconnect via regions VR. In some embodiments, one interconnect via region VR may extend on one side of the device attach region DR and another interconnect via region may extend on an opposite side of the device attach region DR. In some embodiments, the interconnect via regions VR may be separated by the device attach region DR. In some alternative embodiments, the interconnect via region VR may have an annular shape encircling the device attach region DR.

Referring to FIG. **1**D, in some embodiments contact pads **410** are formed in the openings **330** and **340**. In some embodiments, the contact pads **410** include a base layer **410***a* formed in the openings **330** and **340** and contacting the metallization tier **320**, and a bonding layer **410***b* stacked on the base layer **410***a*. In some embodiments, a material of the base layer **410***a* may include mostly copper, possibly in combination with other metals, and a material for the bonding layer **410***b* includes copper. In some embodiments, a seed layer (not shown) may be formed before the base layer **410***a*, and be disposed in between the base layer **410***a* and the metallization tier **320** or the dielectric layer **310**. The base layer **410***a* may be plated on the seed layer, and the bonding layer **410***b* may be plated on the base layer **410***a*. In some embodiments, the contact pads **410** are electrically connected with the underlying semiconductor die **100** or **110**. That is, the contact pads **410** overlying the semiconductor die **100** may be connected to the semiconductor die **100** (but not the semiconductor die **110**), and the contact pads **410** overlying the semiconductor die **110** may be connected to the semiconductor die **110** (but not the semiconductor die **100**). That is, the inner redistribution layer **300** may not directly interconnect the semiconductor dies **100** and **110**.

Referring to FIG. **1**E, in some embodiments a blanket photoresist layer BPR**1** is disposed on the inner redistribution layer **300**. The thickness TBPR**1** of the blanket photoresist layer BPR**1** may be such that the contact pads **410** are completely covered by the blanket photoresist layer BPR**1**, and the blanket photoresist layer BPR**1** may extend over the contact pads **410**. In some embodiments, the thickness TBPR**1** of the blanket photoresist layer BPR**1** measured from the top surface **310***t* of the dielectric layer **310** to the top surface BPR**1***t* of the blanket photoresist layer BPR**1** may be in the range from 3 micrometers to 10 micrometers. In some embodiments, the thickness TBPR**1** may be from 10 to 15 times the protruding height H**410** of the contact pads **410** over the inner redistribution layer **300**. The protruding height H**410** may be considered as the distance between the level height reached by the contact pads **410** and the top surface **310***t* of the dielectric layer **310**. In some embodiments, the blanket photoresist layer BPR**1** includes a negative photoresist or a positive photoresist. In some embodiments, the blanket photoresist layer BPR**1** is laminated as a dry film on the inner redistribution layer **300**.

Referring to FIG. **1**E and FIG. **1**F, the blanket photoresist layer BPR**1** may be patterned (for example, via exposure and development steps) to form a photoresist layer PR**1** including openings OP**1** exposing the contact pads **410** in the interconnect via region(s) VR. Referring to FIG. **1**F and FIG. **1**G, a conductive material **420***a* is disposed in the openings OP**1** on the contact pads **410**. In some embodiments, the conductive material **420***a* may include copper, nickel, tin, palladium, gold, titanium, aluminum, or alloys thereof. In some embodiments, the conductive material **420***a* may be formed by a plating process. The plating process may be, for example, electro-plating, electroless-plating, immersion plating, or the like. In some embodiments, the conductive material **420***a* may be deposited on a seed layer (not shown) provided before the photoresist layer PR**1**. In some embodiments, the formation of the seed layer may be skipped, as the contact pads **410** can seed the deposition of the conductive material **420***a*.

In some embodiments, referring to FIG. **1**G and FIG. **1**H, the photoresist layer PR**1** may be removed or stripped through, for example, etching, ashing, or other suitable removal processes. In some embodiments, the conductive material **420***a* located in the openings OP**1** forms the through interconnection vias (TIVs) **420**. Upon removal of the photoresist layer PR**1**, portions of seed layer not covered by the TIVs **420** are removed, for example, through an etching process. In some embodiments, the material of the TIVs **420** is different from the material of the seed layer, so the portions of the seed layer exposed after removal of the photoresist layer PR**1** may be removed through selective etching. However, the disclosure is not limited thereto. In some alternative embodiments, other suitable methods may be utilized to form the TIVs **420**. For example, pre-fabricated TIVs **420** (e.g., pre-fabricated conductive pillars) may be picked-and-placed onto the inner redistribution layer **300**.

Referring to FIG. **1**I, in some embodiments, a semiconductor bridge **430** and integrated passive devices **440** are connected to the contact pads **410** in between the TIVs **420**, in the device attach region DR. The semiconductor bridge **430** is bonded to some of the contact pads **410** electrically connected to the semiconductor die **100** and to some of the contact pads **410** electrically connected to the semiconductor die **110**, so to establish electrical connection between the semiconductor dies **100** and **110**. As illustrated in FIG. **1**I, in some embodiments, the semiconductor bridge **430** includes a semiconductor substrate **431**, a dielectric layer **432** disposed at a front surface **430***f* of the semiconductor bridge **430**, and interconnection conductive patterns **433** embedded in the dielectric layer **432** and the semiconductor substrate **431**. The semiconductor substrate **431** may be made of suitable semiconductor materials, similar to what was previously discussed for the semiconductor substrates of the semiconductor dies **100**, **110**. The interconnection conductive patterns **433** are in electrical contact with conductive terminals **434** formed on the dielectric layer **432** at the front surface **430***f* of the semiconductor bridge **430**. The conductive terminals **434** may be micro-bumps. For example, the conductive terminals **434** may include a conductive post **434***a* and a solder cap **434***b* disposed on the conductive post **434***a*. In some embodiments, the conductive posts **434***a* may be copper posts. However, the disclosure is not limited thereto, and other conductive structures such as solder bumps, gold bumps or metallic bumps may also be used as the conductive terminals **434**. In some embodiments, the semiconductor bridge **430** is disposed with the front surface **430***f* directed towards the semiconductor dies **100**, **110**, so that the conductive terminals **434** can be bonded to the

contact pads **410**. That is, the solder cap **434***b* may be disposed on the bonding layer **410***b* of the contact pads **410** to establish electrical connection. In some embodiments, the interconnection conductive patterns **433** of the semiconductor bridge **430** electrically connect the semiconductor dies **100** and **110** through the intervening contact pads **410**. The conductive terminals **434** may be bonded to the contact pads **410** through a reflow process. Upon bonding the semiconductor bridge **430** to the contact pads **410**, electrical connection between the semiconductor dies **100** and **110** is established through the inner redistribution layer **300**, the contact pads **410**, the conductive terminals **434** and the interconnection conductive patterns **433**. In some embodiments, the inner redistribution layer **300** does not directly interconnect the semiconductor dies **100**, **110**. In some embodiments, the semiconductor bridge **430** connects at least one contact pad **410** electrically connected to the semiconductor die **100** to another contact pad **410** connected to the semiconductor die **110**. In some embodiments, the semiconductor bridge **430** connects one or more contact pads **410** overlying the semiconductor die **100** with one or more contact pads **410** overlying the semiconductor die **110**. In some embodiments, where a gap exists between adjacent semiconductor dies **100**, **110**, the semiconductor bridge **430** extends over such gap. In some embodiments, the semiconductor bridge **430** functions as an interconnecting structure for adjacent semiconductor dies **100**, **110** and provides shorter electrical connection paths between the adjacent semiconductor dies **100**, **110**.

In some embodiments, the integrated passive devices **440** are provided on the contact pads **410** not occupied by the semiconductor bridge **430** in the device attach region DR. In some embodiments, the integrated passive devices **440** are placed over the carrier C through a pick-and-place method. Even though only two integrated passive devices **440** are presented in FIG. **1**I within a package unit PU for illustrative purposes, the disclosure is not limited by the number of integrated passive devices **440** included in a package unit PU. In some embodiments, an individual integrated passive device **440** includes a semiconductor substrate **441**, a dielectric layer **442** formed on the semiconductor substrate **441** at a front surface **440***f* of the integrated passive device **440**, and conductive terminals **443** formed on the front surface **440***f*. The semiconductor substrate **441** may be made of suitable semiconductor materials, similar to what was previously discussed for the semiconductor substrates of the semiconductor dies **100**, **110**. In some embodiments, the integrated passive devices **440** are chips of integrated passive devices and function as capacitors, inductors, resistors, or the like. In some embodiments, each integrated passive device **440** may independently function as a capacitor having different capacitance values, resonance frequencies, and/or different sizes, an inductor, or the like. In some embodiments the conductive terminals **443** may be micro-bumps. For example, the conductive terminals **443** may include a conductive post **443***a* and a solder cap **443***b* disposed on the conductive post **443***a*. In some embodiments, the conductive posts **443***a* may be copper posts. However, the disclosure is not limited thereto, and other conductive structures such as solder bumps, gold bumps or metallic bumps may also be used as the conductive terminals **443**. In some embodiments, the integrated passive devices **440** are disposed with the front surfaces **440***f* directed towards the semiconductor dies **100**, **110**, so that the conductive terminals **443** can be bonded to the contact pads **410**. That is, the solder caps **443***b* may be disposed on the contact pads **410** to establish electrical connection with the semiconductor dies **100**, **110**. In some embodiments, the rear surfaces **430***r*, **440***r* of the semiconductor bridge **430** and the integrated passive devices **440** (the surfaces **430***r*, **440***r* opposite to the corresponding front surfaces **430***f*, **440***f*) may be located at a level height higher than the level height reached by the TIVs **420** with respect to the inner redistribution layer **300**.

Referring to FIG. **1**J, in some embodiments, an encapsulant **450** is formed on the inner redistribution layer **300**, encapsulating the contact pads **410**, the TIVs **420**, the semiconductor bridge **430** and the integrated passive devices **440**. In some embodiments, the encapsulant **450** is formed through an over-molding process, for example, through a compression molding process. The encapsulant **450** may initially cover the TIVs **420** and the rear surfaces **430***r*, **440***r* of the semiconductor bridge **430** and the integrated passive devices **440**, respectively. In some embodiments, similar materials as the ones listed above for the encapsulant **200** may be used for the encapsulant **450**. Referring to FIG. **1**J and FIG. **1**K, the encapsulant **450** may be thinned until the TIVs **420** are exposed. For example, a planarization process may be performed removing portions of the encapsulant **450** and, if needed, of the semiconductor substrates **431**, **441** of the semiconductor bridge **430** and the integrated passive devices **440** from the side of the rear surfaces **430***r*, **440***r*. In some embodiments, the planarization of the encapsulant **450** includes performing a mechanical grinding process and/or a chemical mechanical polishing (CMP) process. Following planarization, top surfaces **420***t* of the TIVs **420**, the rear surface **430***r* of the semiconductor bridge **430**, the rear surfaces **440***r* of the integrated passive devices **440**, and the top surface **450***t* of the encapsulant **450** may be substantially flush with respect to each other (be at substantially the same level height, coplanar with respect to each other). In some embodiments, the contact pads **410**, the TIVs **420**, the semiconductor bridge **430**, the integrated passive devices **440**, and the encapsulant **450** are considered parts of a bridging layer **400** stacked on the inner redistribution layer **300**. That is, the semiconductor bridge **430** and the integrated passive devices **440** are embedded in the bridging layer **400**.

Referring to FIG. **1**L, an outer redistribution layer **500** is formed on the bridging layer **400**. The outer redistribution layer **500** includes dielectric layers **510**, one or more metallization tiers **520**, and, optionally, under-bump metallurgies **530**. The outer redistribution layer **500** may have a similar structure and be formed following similar processes as the ones previously described for the inner redistribution layer **300**, and a detailed description thereof is omitted herein. In some embodiments, the (outermost) dielectric layer **510** is patterned to expose the underlying metallization tier **520**. The under-bump metallurgies **530** are optionally conformally formed in the openings of the (outermost) dielectric layer **510** exposing the metallization tier **520**, and may further extend over portions of the exposed surface of the (outermost) dielectric layer **510**. In some embodiments, the under-bump metallurgies **530** include multiple stacked layers of conductive materials. For example, the under-bump metallurgies **530** may include one or more metallic layers stacked on a seed layer. In some embodiments, the inner redistribution layer **300**, the bridging layer **400**, and the outer redistribution layer **500** may collectively form a redistribution structure **600**, having embedded therein the semiconductor bridge **430** and the integrated passive devices **440**.

Connective terminals **700** are formed on the redistribution structure **600** on an opposite side with respect to the semiconductor dies **100**, **110**, on the under-bump metallurgies

530 (if included) of the outer redistribution layer 500. In some embodiments, the connective terminals 700 are formed on the under-bump metallurgies 530, and are connected to the TIVs 420 and the semiconductor die(s) 100, 110 via the metallization tiers 520 and 320. In some embodiments, the connective terminals 700 are attached to the under-bump metallurgies 530 through a solder flux. In some embodiments, the connective terminals 700 are controlled collapse chip connection (C4) bumps. In some embodiments, the connective terminals 700 include a conductive material with low resistivity, such as Sn, Pb, Ag, Cu, Ni, Bi, or an alloy thereof.

In some embodiments, referring to FIG. 1L and FIG. 1M, a singulation step is performed to separate the individual semiconductor packages P10, for example, by cutting through the reconstructed wafer RW along the scribe lanes SC arranged between individual package units PU. In some embodiments, the singulation process typically involves performing a wafer dicing process with a rotating blade and/or a laser beam. In some embodiments, the carrier C is separated from the semiconductor packages SP10 following singulation. When the de-bonding layer (e.g., the LTHC release layer) is included, the de-bonding layer may be irradiated with a UV laser so that the carrier C and the de-bonding layer are easily peeled off from the semiconductor packages SP10. Nevertheless, the de-bonding process is not limited thereto, and other suitable de-bonding methods may be used in some alternative embodiments.

FIG. 1M is a schematic cross-sectional view of the semiconductor package SP10 according to some embodiments of the disclosure. The semiconductor package SP10 may include the semiconductor dies 100, 110, the encapsulant 200, the redistribution structure 600 including the redistribution layers 300 and 500 and the bridging layer 400, and the connective terminals 700. In some embodiments, the inner redistribution layer 300 is formed on the semiconductor dies 100, 110 and the encapsulant 200, the bridging layer 400 is formed on the inner redistribution layer 300, and the outer redistribution layer 500 is formed on the bridging layer 400. In some embodiments, the inner redistribution layer 300 and the outer redistribution layer 500 are connected by the TIVs 420 of the bridging layer 400. The TIVs 420 may be formed in the interconnect via region VR of the semiconductor package P10. The semiconductor package P10 further includes a device attach region DR, in which the semiconductor bridge 430 and the integrated passive devices 440 are disposed within the bridging layer 400. That is, the semiconductor bridge 430 and the integrated passive devices 440 are buried (embedded) within the bridging layer 400. That is, the integrated passive devices 440 are closer to the semiconductor dies 100, 110 with respect to the connective terminals 700. In some embodiments, by burying the integrated passive devices 440 within the bridging layer 400, a connection distance between the integrated passive devices 440 and the semiconductor dies 100, 110 may be shortened, simplifying the routing of the redistribution structure 600 and increasing the efficiency of the semiconductor package P10. In some embodiments, the increase in efficiency may be achieved without substantially increasing the thickness of the semiconductor package P10.

As illustrated in FIG. 1N, in some embodiments the connective terminals 700 may be used to integrate the semiconductor package P10 in larger electronic devices (e.g., the electronic device D10). For example, the semiconductor package P10 may be connected to a circuit substrate 800 such as a mother board, a printed circuit board, or the like, via the connective terminals 700.

FIG. 2A through FIG. 2F are schematic cross-sectional views of structures formed during a manufacturing process of a semiconductor package P20 according to some embodiments of the disclosure. In some embodiments, the structure of FIG. 2A may be formed following similar steps as the ones previously described with reference to FIG. 1A through FIG. 1E, so that the process will only be briefly outlined here. In some embodiments, the semiconductor dies 100, 110 are disposed on the carrier C and encapsulated in the encapsulant 200, and the inner redistribution layer 1300 is formed on the semiconductor dies 100, 110 and the encapsulant 200. The inner redistribution layer 1300 includes the dielectric layers 1310 and one or more metallization tiers 1320 electrically connected to the contact pads 104, 114 of the semiconductor dies 100, 110. A difference with respect to the previous process is that the blanket photoresist layer BPR2 is laminated on the inner redistribution layer 1300 without forming the contact pads (e.g., the contact pads 410 of FIG. 1D). That is, the blanket photoresist layer BPR2 may fill the openings 1330 and 1340. Referring to FIG. 2A and FIG. 2B, the blanket photoresist layer BPR2 is patterned to form the photoresist layer PR2 including openings OP2 which expose the openings 1340 in the interconnect via region VR. Thereafter, as illustrated in FIG. 2C and FIG. 2D, the openings OP2 and 1340 are filled with the conductive material 1420*a* and the photoresist layer PR2 is stripped to expose again the inner redistribution layer 1300. The openings 1330 of the inner redistribution layer 1300 expose portions of the metallization tier 1320 in the device attach region DR. Similar to what was previously described, some of the openings 1330 overlay the semiconductor die 100 and some other openings 1330 overlay the semiconductor die 110. Referring to FIG. 2E, the semiconductor bridge 1430 and the integrated passive devices 1440 are disposed on the inner redistribution layer 1300 in between the TIVs 1420, with the corresponding front surfaces 1430*f*, 1440*f* directed towards the semiconductor dies 100, 110. A difference with respect to the process previously described is that the semiconductor bridge 1430 and the integrated passive devices 1440 are directly connected to the metallization tier 1320, without intervening contact pads 410 (illustrated, e.g., in FIG. 1I) formed therebetween. That is, the conductive terminals 1434 of the semiconductor bridge 1430 and the conductive terminals 443 of the integrated passive devices 440 directly contact the metallization tier 1320. After the semiconductor bridge 1430 and the integrated passive devices 1440 are bonded to the inner redistribution layer 1300, process steps similar to the ones described before with reference to FIG. 1J to FIG. 1M result in the formation of the semiconductor package P20 illustrated in FIG. 2F. In some embodiments, a difference between the semiconductor package P20 illustrated in FIG. 2F and the semiconductor package P10 illustrated in FIG. 1M lies in the absence of contact pads 410 in the bridging layer 1400. That is, the bridging layer 1400 of the redistribution structure 1600 of the semiconductor package P20 includes the TIVs 1420, the semiconductor bridge 1430, and the integrated passive devices 1440 encapsulated by the encapsulant 1450, with the semiconductor bridge and the integrated passive devices 1440 directly connected to the inner redistribution layer 1300. Other features may be common between the two semiconductor packages P10 and P20. It should be noted that for every embodiment disclosed in which the bridging layer does not include the contact pads like the contact pads 410 of FIG. 1D, there is a corresponding alternative embodiment in which such contact pads are included. For the sake of brevity, in the rest of the disclosure will be illustrated only

structures in which the contact pads are not included in the bridging layer. Also, features not explicitly addressed of the semiconductor packages described hereafter may be considered to be similar to the features previously discussed for the semiconductor packages P**10** and P**20**.

FIG. **3**A through FIG. **3**D are schematic cross-sectional views of enlarged portions of semiconductor packages according to some embodiments of the disclosure. The views of FIG. **3**A through FIG. **3**D may correspond to the area A illustrated in FIG. **2**F. In FIG. **3**A is illustrated an enlarged portion of a semiconductor package P**30** according to some embodiments of the disclosure. In the semiconductor package P**30**, the semiconductor bridge **430** includes nested interconnection conductive patterns **1433** connected to the conductive terminals **1434** and interconnecting the semiconductor dies **100**, **110**. In some embodiments, one end of an interconnection conductive pattern **1433** is connected to a conductive terminal **1434** overlying and electrically connected to the semiconductor die **100**, and the other end of the interconnection conductive pattern **1433** is connected to a conductive terminal **1434** overlying and electrically connected to the semiconductor die **110**. In some embodiments, the integrated passive device **1440** illustrated in FIG. **3**A overlays the semiconductor die **110**, and includes passive device patterns **1444** (e.g., inductor patterns) formed in the semiconductor substrate **1441**. In some embodiments, the passive device patterns **1444** are connected to the conductive terminals **1443** by conductive patterns **1445** extending through the dielectric layer **1442**. In some embodiments, the integrated passive device **1440** overlying the semiconductor die **100** (illustrated, e.g., in FIG. **2**F) may have a similar structure.

In FIG. **3**B is illustrated an enlarged portion of a semiconductor package P**40** according to some embodiments of the disclosure. A difference between the semiconductor package P**40** and the semiconductor package P**30** of FIG. **3**A lies in the integrated passive device **1440** further including through semiconductor vias (TSVs) **1446** extending through the semiconductor substrate **1441** and establishing dual-side vertical connection between the front surface **1440***f* and the rear surface **1440***r* of the integrated passive device **1440**. In some embodiments, one end of a TSV **1446** is connected (e.g., through intervening conductive patterns **1445**) to a conductive terminal **1443**, while the other end of the TSV **1446** is connected to the outer redistribution layer **1500** (illustrated, e.g., in FIG. **2**F). In some embodiments, by including TSVs **1446** in the integrated passive devices **1440** it may be possible to minimize or avoid loss of I/O sites from the semiconductor dies **100**, **110**.

In FIG. **3**C is illustrated an enlarged portion of a semiconductor package P**50** according to some embodiments of the disclosure. A difference between the semiconductor package P**50** and the semiconductor package P**30** of FIG. **3**A lies in the semiconductor bridge **1430** further including through semiconductor vias (TSVs) **1435** extending through the semiconductor substrate **1431** and establishing dual-side vertical connection between the front surface **1430***f* and the rear surface **1430***r* of the semiconductor bridge **1430**. In some embodiments, one end of a TSV **1435** is connected (e.g., through intervening conductive patterns **1436**) to a conductive terminal **1434**, while the other end of the same TSV **1435** is connected to the outer redistribution layer **1500** (illustrated, e.g., in FIG. **2**F). In some embodiments, by including TSVs **1435** in the semiconductor bridge **1430** it may be possible to minimize or avoid loss of I/O sites from the semiconductor dies **100**, **110**. In some embodiments, the conductive patterns **1436** may also establish electrical connection between the interconnection conductive patterns **1433** and the conductive terminals **1434**.

In FIG. **3**D is illustrated an enlarged portion of a semiconductor package P**60** according to some embodiments of the disclosure. A difference between the semiconductor package P**60** and the semiconductor package P**30** of FIG. **3**A lies in that TSVs **1435**, **1446** are formed both in the integrated passive devices **1440** and the semiconductor bridge **1430** (similar to what was discussed with reference to FIG. **3**B and FIG. **3**C). In some embodiments, by including TSVs **1435**, **1446** in both of the semiconductor bridge **1430** and the integrated passive devices **1440** it may be possible to further minimize or avoid loss of I/O sites from the semiconductor dies **100**, **110**.

In FIG. **4**A through FIG. **4**C are illustrated schematic cross-sectional views of semiconductor packages according to some embodiments of the disclosure. The schematic cross-sectional views of FIG. **4**A through FIG. **4**C may be taken in a plane perpendicular with respect to the view illustrated, e.g., in FIG. **2**F, at a level height corresponding to the line I-I of FIG. **2**F. That is, the schematic cross-sectional views of FIG. **4**A through FIG. **4**C may be taken in planes passing through the bridging layer **1400** illustrated in FIG. **2**F.

In FIG. **4**A is illustrated a schematic cross-sectional view of a semiconductor package P**70** according to some embodiments of the disclosure. In some embodiments, the semiconductor package P**70** includes a device attach region DR and two interconnect via regions VR disposed at opposite sides of the device attach region DR. The semiconductor bridge **1430** is disposed towards the center of the device attach region DR (and of the semiconductor package P**70**), with a first integrated passive device **1440**A on one side and a second integrated passive device **1440**B on an opposite side. An interconnect via region VR may extend from the integrated passive device **1440**A towards the edge of the semiconductor package P**70**, and the other interconnect via region VR may extend from the other integrated passive device **1440**B towards the opposite edge of the semiconductor package P**70**.

In FIG. **4**B is illustrated a schematic cross-sectional view of a semiconductor package P**80** according to some embodiments of the disclosure. A difference between the semiconductor package P**80** and the semiconductor package P**70** of FIG. **4**A lies in the number of integrated passive devices **1440**A-D included. That is, in the semiconductor package P**80**, two integrated passive devices **1440**A, **1440**C are disposed on one side of the semiconductor bridge **1430**, and other two integrated passive devices **1440**B, **1440**D are disposed on the opposite side of the semiconductor bridge **1430**. Of course, the disclosure is not limited by the number of integrated passive devices **1440** included. In some alternative embodiments, more or fewer integrated passive devices **1440** may be included according to design and production requirements. In some embodiments, disposing the integrated passive devices **1440** closer to the semiconductor bridge **1430** (as illustrated, e.g., in FIG. **4**A or FIG. **4**B) may save routing area, increasing the number of available I/O sites.

In FIG. **4**C is illustrated a schematic cross-sectional view of a semiconductor package P**90** according to some embodiments of the disclosure. A difference between the semiconductor package P**90** and the semiconductor package P**80** of FIG. **4**B lies in the fact that the TIVs **1420** are disposed in between the semiconductor bridge **1430** and the integrated passive devices **1440**A-D. That is, the semiconductor package P**90** includes a central device attach region CDR in

which the semiconductor bridge **1430** is disposed, and two peripheral device attach regions PDR in which the integrated passive devices **1440**A-D are disposed. The peripheral device attach regions PDR are disposed at opposite sides of the central device attach region CDR, and are separated from the central device attach region CDR by the interconnect via regions VR. That is, while in the semiconductor package P**70** or P**80** of FIG. **4**A or FIG. **4**B the integrated passive devices **1440** are disposed in between the semiconductor bridge **1430** and the TIVs **1420**, in the semiconductor package P**90** the TIVs **1420** are disposed in between the semiconductor bridge **1430** and the integrated passive devices **1440**. However, the disclosure is not limited thereto. In some alternative embodiments, the integrated passive devices **1440** may be included also in the central device attach region CDR as well as in the peripheral device attach regions PDR.

FIG. **5** is a schematic cross-sectional view of a semiconductor package P**100** according to some embodiments of the disclosure. A difference between the semiconductor package P**100** and the semiconductor package P**20** of FIG. **2**F lies in the integrated passive devices being formed within the semiconductor bridge **2430**. That is, in the semiconductor package P**100**, the bridging layer **2400** of the redistribution structure **2600** may not include separate integrated passive devices (e.g., the integrated passive devices **1440** illustrated in FIG. **2**F). Rather, passive devices patterns **2437** (e.g., the deep trench capacitors, inductors, etc.) are formed in the semiconductor substrate **2431** of the semiconductor bridge **2430** beside the interconnection conductive patterns **2433** interconnecting the semiconductor dies **100**, **110**. The passive device patterns **2437** may be electrically connected to the semiconductor dies **100**, **110** through some dedicated conductive terminals **2434** of the semiconductor bridge **2430**. In some embodiments, the integrated passive devices may be formed at opposite sides of the interconnection conductive patterns **2433**. For example, a first passive device pattern **2437** may be formed on one side of the interconnection conductive patterns **2433** to overlay and be connected with the semiconductor die **100**, and another first device pattern **2437** may be formed on an opposite side of the interconnection conductive patterns **2433** to overlay and be connected with the semiconductor die **110**.

FIG. **6** is a schematic cross-sectional view of a semiconductor package P**110** according to some embodiments of the disclosure. A difference between the semiconductor package P**110** and the semiconductor package P**20** of FIG. **2**F lies in position and orientation of the integrated passive devices **3440** within the bridging layer **3400** of the redistribution structure **3600**. In some embodiments, the integrated passive devices **3440** are stacked on the semiconductor bridge **3430** rather than being disposed side by side with respect to the semiconductor bridge **3430** (as illustrated, for example, in FIG. **2**F). That is, the semiconductor bridge **3430** may be disposed between the integrated passive devices **3440** and the inner redistribution layer **3300**, and the integrated passive devices **3440** may be disposed between the semiconductor bridge **3430** and the outer redistribution layer **3500**. The integrated passive devices **3440** and the semiconductor bridge **3430** may be disposed in a back-to-back configuration. That is, the rear surfaces **3440***r* of the integrated passive devices **3440** may be directed towards the rear surface **3430***r* of the semiconductor bridge **3430** (and the top surfaces **100***t*, **110***t* of the semiconductor dies **100**, **110**). The front surface **3430***f* of the semiconductor bridge **3430** (the contact surface) is directed towards the inner redistribution layer **3300**, so that the semiconductor bridge **3430** is directly connected to the inner redistribution layer **3300**, and the front surfaces **3440***f* of the integrated passive devices **3440** (the contact surfaces) are directed towards the outer redistribution layer **3500**, so that the integrated passive devices **3440** are directly connected to the outer redistribution layer **3500**. The integrated passive devices **3440** may be connected to the semiconductor dies **100**, **110** via (in order) the outer redistribution layer **3500**, the TIVs **3420** and the inner redistribution layer **3300**. In some embodiments, the integrated passive devices **3440** may be secured to the semiconductor bridge **3430** via portions of die attach film **3460**. For example, strips of die attach film **3460** may be disposed in between the rear surface **3430***r* of the semiconductor bridge and the rear surfaces **3440***r* of the integrated passive devices **3440**. Portions of the encapsulant **3450** may extend in between the integrated passive devices **3440** over the semiconductor bridge **3430**. That is, the encapsulant **3450** may separate the semiconductor bridge **3430** from the outer redistribution layer **3500**.

FIG. **7** is a schematic cross-sectional view of a semiconductor package P**120** according to some embodiments of the disclosure. A difference between the semiconductor package P**120** and the semiconductor package P**110** of FIG. **6** lies in the configuration of the integrated passive devices **4440** with respect to the semiconductor bridge **4430**. That is, in the semiconductor package P**120**, the integrated passive devices **4440** are disposed in a face-to-back configuration with respect to the semiconductor bridge **4430**, with the respective front surfaces **4440***f* directed towards the rear surface **4430***r* of the semiconductor bridge **4430**. As such, the rear surfaces **4440***r* of the integrated passive devices **4440** are directed towards the outer redistribution layer **4500**. In the semiconductor package P**120**, the integrated passive devices **4440** may be connected to the semiconductor dies **100**, **110** through the semiconductor bridge **4430**. For example, the semiconductor bridge **4430** includes through semiconductor vias (TSVs) **4435** and conductive patterns **4436** establishing dual-side vertical electrical connection between the front surface **4430***f* and the rear surface **4430***r*. The passive device patterns **4444** formed in the integrated passive devices **4440** may be connected to the TSVs **4435** either directly, or through intervening conductive terminals **4443**. As such, the integrated passive devices **4440** may be connected to the semiconductor dies **100**, **110** via the semiconductor bridge **4430** and the inner redistribution layer **4300**.

FIG. **8**A through FIG. **8**C are schematic cross-sectional views of structures formed during a manufacturing process of a semiconductor package P**130** according to some embodiments of the disclosure. Referring to FIG. **8**A, TIVs **5050**, semiconductor dies **5100**, and integrated passive devices **5440** are provided over the carrier C. In some embodiments, the order of providing the TIVs **5050**, the semiconductor dies **5100** and the integrated passive devices **5440** may be selected according to process requirements. The TIVs **5050** may be provided, for example, through a photolithography process, a plating process, a photoresist stripping process, and/or any other suitable process. In some embodiments, the TIVs **5050** may be pre-fabricated conductive pillars which are picked and placed over the carrier C. The semiconductor dies **5100** and the integrated passive devices **5440** are respectively similar to the semiconductor dies **100**, **110** and the integrated passive devices **440** (illustrated, e.g., illustrated in FIG. **1**I), and a detailed description thereof is omitted. The semiconductor dies **5100** and the integrated passive devices **5440** may be disposed on the carrier C through a pick-and-place process, for example. In some embodiments, the semiconductor dies **5100** and the

integrated passive devices **5440** are disposed in between the TIVs **5050** in a face-up configuration, with the respective top surfaces **5100***t* and front surfaces **5440***f* (contact surfaces) facing away from the carrier C.

Referring to FIG. **8**B and FIG. **8**C, the TIVs **5050**, the semiconductor dies **5100**, and the integrated passive devices **5440** are molded in the encapsulant **5200**, and the redistribution structure **5300** and the connective terminals **5400** are subsequently provided thereon. The encapsulant **5200** may be produced employing similar materials and following similar processes as the encapsulant **200** (illustrated, e.g., in FIG. **1**B). The redistribution structure **5300** may be produced following similar steps as previously described with respect to the redistribution layers **300** and **500** (illustrated, e.g., in FIG. **1**L). The redistribution structure **5300** includes a single redistribution layer including the dielectric layers **5310**, one or more metallization tiers **5320**, and, optionally, the under-bump metallurgies **5330** disposed in contact with the (outermost) metallization tier **5320**.

In some embodiments, the semiconductor package P**130** may be provided after singulation and removal of the carrier C. As illustrated in FIG. **8**C, a difference between the semiconductor package P**130** and the semiconductor package P**20** of FIG. **2**F lies in the position of the integrated passive devices **5440**. That is, in the semiconductor package P**130**, the integrated passive devices **5440** are embedded in the encapsulant **5200** beside the semiconductor die(s) **5100**, rather than in a bridging layer of the redistribution structure **5300**. More specifically, the integrated passive devices **5440** are disposed at the sides of the semiconductor die(s) **5100**, with the respective front surfaces **5440***f* and top surface(s) **5100***t* (contact surfaces) being substantially coplanar. In some embodiments, the semiconductor package P**130** does not include a semiconductor bridge, and the redistribution structure **5300** does not include a bridging layer. Rather, as illustrated in the inset of FIG. **8**C, the semiconductor die(s) **5100** may be connected with each other or with the integrated passive devices **5440** by conductive patterns of some metallization tiers **5320** of the redistribution structure **5300**. By doing so, in some embodiments, the connection between the integrated passive devices **5440** and the semiconductor die(s) **5100** may be shortened, thus increasing the efficiency of the semiconductor package P**130**.

In some embodiments, the TIVs **5050** are available to establish dual-side vertical electrical connection. For example, as illustrated in FIG. **8**D, the semiconductor package P**130** may be adapted to be a bottom package BP of a package-on-package structure, as in the semiconductor package P**135**. An upper package UP may be connected to the bottom package BP through connective terminals **5500** which contact first ends of the TIVs **5050**, while the other ends are connected to the redistribution structure **5300**. In some embodiments, the upper package UP may be a memory package. For example, the upper package UP may include a chip stack **5610** wire bonded through conductive wires **5620** to a redistribution layer **5700**. The redistribution layer **5700** may connect the conductive wires **5620** and the chip stack **5610** to the connective terminals **5500**. In some embodiments, the chip stack **5610** and the conductive wires **5620** may be encapsulated by an encapsulant **5800**. An underfill **5900** may be provided between the bottom package BP and the upper package UP to protect the connective terminals **5500** from thermal and mechanical stresses, thus securing the connection within the semiconductor package P**135**. However, the disclosure is not limited thereto. In some embodiments, other types of packages, including different components or having different functions, may be used as the upper package UP.

FIG. **9** is a schematic cross-sectional view of a semiconductor package P**145** according to some embodiments of the disclosure. The semiconductor package P**145** may be similar to the semiconductor package P**135** of FIG. **8**D. A difference between the semiconductor package P**145** and the semiconductor package P**135** of FIG. **8**D lies in the use of the semiconductor package P**140** as the bottom package BP. In the semiconductor package P**140**, the redistribution structure **6600** includes an inner redistribution layer **6300**, a bridging layer **6400**, and an outer redistribution layer **6500** stacked in sequence, similar to the semiconductor package P**20** of FIG. **2**F. However, in the semiconductor package P**140**, only the semiconductor bridge **6430** is embedded in the bridging layer **6400**, while the integrated passive devices **6440** are embedded in the encapsulant **6200** at the sides of the semiconductor dies **6100**, **6110**. In some embodiments, the semiconductor bridge **6430** interconnects the semiconductor dies **6100**, **6110**. The inner redistribution layer **6300** may interconnect the semiconductor dies **6100**, **6110** with adjacent integrated passive devices **6440**. However, the disclosure is not limited thereto. In some alternative embodiments, the semiconductor bridge **6430** may also interconnect the semiconductor dies **6100**, **6110** with adjacent integrated passive devices **6440**.

FIG. **10** is a schematic cross-sectional view of a semiconductor package P**150** according to some embodiments of the disclosure. A difference between the semiconductor package P**150** and the semiconductor package P**140** of FIG. **9** lies in the position of the integrated passive devices **7440**. That is, in the semiconductor package P**150**, the integrated passive devices **7440** are stacked on the semiconductor dies **7100**, **7110**, and are disposed in between the semiconductor dies **7100**, **7110** and the inner redistribution layer **7300** of the redistribution structure **7600**. In some embodiments, the integrated passive devices **7440** are bonded to the semiconductor dies **7100**, **7110** in a face-to-face configuration, either directly or through intervening conductive terminals **7443**. TIVs **7080** are provided on the semiconductor dies **7100**, **7110** to establish connection between the semiconductor dies **7100**, **7110** and the inner redistribution layer **7300**. The encapsulant **7200** may embed the semiconductor dies **7100**, **7110**, the integrated passive devices **7440**, and the TIVs **7080**. Portions of the encapsulant **7200** may extend between the semiconductor dies **7100**, **7110** and the inner redistribution layer **7300**.

In accordance with some embodiments of the disclosure, a semiconductor package includes a first semiconductor die, a second semiconductor die, a semiconductor bridge, an integrated passive device, a first redistribution layer, and connective terminals. The second semiconductor die is disposed beside the first semiconductor die. The semiconductor bridge electrically connects the first semiconductor die with the second semiconductor die. The integrated passive device is electrically connected to the first semiconductor die. The first redistribution layer is disposed over the semiconductor bridge. The connective terminals are disposed on the first redistribution layer, on an opposite side with respect to the semiconductor bridge. The first redistribution layer is interposed between the integrated passive device and the connective terminals.

In accordance with some embodiments of the disclosure, a semiconductor package includes semiconductor dies, an encapsulant and a redistribution structure. The encapsulant encapsulates the semiconductor dies. The redistribution

structure is disposed on the encapsulant and the semiconductor dies. The redistribution structure includes, in order, a first redistribution layer, a bridging layer, and a second redistribution layer. The bridging layer includes a semiconductor bridge and an integrated passive device. The semiconductor bridge interconnects at least two semiconductor dies of the semiconductor dies. The integrated passive device is electrically connected to at least one of the at least two semiconductor dies through the first redistribution layer.

In accordance with some embodiments of the disclosure, a manufacturing method of a semiconductor package includes the following steps. Semiconductor dies are provided. An encapsulant is formed encapsulating the semiconductor dies. A semiconductor bridge is provided electrically connecting at least two semiconductor dies of the semiconductor dies. An integrated passive device is provided. The integrated passive device is electrically connected to one semiconductor die of the at least two semiconductor dies. A redistribution layer is formed over the integrated passive device and the semiconductor dies.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor package, comprising:

a first redistribution layer;

a semiconductor die, disposed over the first redistribution layer, and having a first surface and a second surface opposite to the first surface;

an integrated passive die, disposed between the first surface of the semiconductor die and the first redistribution layer, and electrically connected to the semiconductor die;

a first encapsulant, formed between first surface of the semiconductor die and the first redistribution layer, and encapsulating the integrated passive die;

connective terminals, deployed at a surface of the first redistribution layer opposite to the first encapsulant; and

a second redistribution layer, extending along the first surface of the semiconductor die, wherein the integrated passive die is in contact with the semiconductor die through the second redistribution layer,

wherein a shortest distance between the integrated passive die and the semiconductor die is shorter than a shortest distance between the first redistribution layer and the semiconductor die, and wherein a front side of the integrated passive die at which micro-bumps are disposed faces toward the first surface of the semiconductor die, and a back side of the integrated passive die is in direct contact with the first redistribution layer, and wherein contact pads are disposed on the second redistribution layer and protruded from the second redistribution layer, through interconnection vias stand on a first group of the contact pads, and the integrated passive die is attached onto a second group of the contact pads via the micro-bumps.

2. The semiconductor package according to claim **1**, wherein the first encapsulant is in contact with the first and second redistribution layers by opposite sides, and through interconnection vias penetrating through the first encapsulant establish conduction paths between the first and second redistribution layers.

3. The semiconductor package according to claim **1**, wherein the integrated passive die comprises through semiconductor vias configured to establish conduction paths between a front side and a back side of the integrated passive die.

4. The semiconductor package according to claim **1**, wherein the integrated passive die is a capacitor die.

5. The semiconductor package according to claim **1**, wherein the micro-bumps include a conductive post and a solder cap disposed on the conductive post.

6. The semiconductor package according to claim **1**, wherein a sidewall of the first redistribution layer, a sidewall of the first encapsulant and a sidewall of the second redistribution layer are substantially aligned with each other.

7. The semiconductor package according to claim **1**, further comprising:

a second encapsulant, disposed over the first encapsulant, and encapsulating the semiconductor die.

8. The semiconductor package according to claim **7**, wherein a sidewall of the first redistribution layer, a sidewall of the first encapsulant and a sidewall of the second encapsulant are substantially aligned with each other.

9. A semiconductor package, comprising:

a first redistribution layer;

an integrated passive die, disposed on the first redistribution layer;

a semiconductor die, stacked over the integrated passive die, and electrically connected to the integrated passive die;

a first encapsulant, formed between the semiconductor die and the first redistribution layer, and encapsulating the integrated passive die;

through interconnection vias, penetrating through the first encapsulant, and located around the integrated passive die, wherein the semiconductor die is electrically connected to the first redistribution layer via the through interconnection vias; and

a second redistribution layer, extending under the semiconductor die, wherein the integrated passive die is in contact with the semiconductor die through the second redistribution layer,

and wherein a shortest distance between the integrated passive die and the semiconductor die is shorter than a shortest distance between the first redistribution layer and the semiconductor die, and wherein a front side of the integrated passive die at which micro-bumps are disposed faces toward the semiconductor die, and a back side of the integrated passive die is in direct contact with the first redistribution layer, and wherein contact pads are disposed on the second redistribution layer and protruded from the second redistribution layer, the through interconnection vias stand on a first group of the contact pads, and the integrated passive die is attached onto a second group of the contact pads via the micro-bumps.

10. The semiconductor package according to claim **9**, wherein the integrated passive die is vertically spaced apart from the semiconductor die.

11. The semiconductor package according to claim **9**, further comprising:

connective terminals, deployed at a surface of the first redistribution layer opposite to the first encapsulant, wherein the integrated passive die and the connective terminals are disposed at opposite sides of the first redistribution layer.

12. The semiconductor package according to claim **9**, wherein the semiconductor die is electrically connected to the integrated passive die and the first redistribution layer through the second redistribution layer.

13. The semiconductor package according to claim **12**, wherein the integrated passive die, the first encapsulant and the through interconnection vias are located between the first and second redistribution layers.

14. The semiconductor package according to claim **12**, wherein a sidewall of the first redistribution layer, a sidewall of the first encapsulant and a sidewall of the second redistribution layer are substantially aligned with each other.

15. A semiconductor package, comprising:

a first redistribution layer;

a semiconductor die, disposed over the first redistribution layer, and having a first surface and a second surface opposite to the first surface;

an integrated passive die, attached to the first surface of the semiconductor die, and located between the semiconductor die and the first redistribution layer;

a first encapsulant, encapsulating the semiconductor die and the integrated passive die;

a second redistribution layer, lying below the first redistribution layer; and

a second encapsulant, vertically separating the second redistribution layer from the first redistribution layer,

wherein a shortest distance between the integrated passive die and the semiconductor die is shorter than a shortest distance between the first redistribution layer and the semiconductor die.

16. The semiconductor package according to claim **15**, further comprising:

second through interconnection vias, penetrating through the second encapsulant to establish conduction paths between the first and second redistribution layers.

17. The semiconductor package according to claim **15**, further comprising:

connective terminals, deployed at a surface of the second redistribution layer facing away from the second encapsulant.

18. The semiconductor package according to claim **15**, wherein a sidewall of the first redistribution layer, a sidewall of the first encapsulant, a sidewall of the second redistribution layer and a sidewall of the second encapsulant are substantially aligned with each other.

19. The semiconductor package according to claim **15**, further comprising:

first through interconnection vias, penetrating through the first encapsulant to establish conduction paths between the semiconductor die and the first redistribution layer.

20. The semiconductor package according to claim **19**, wherein the integrated passive die and the first through interconnection vias are overlapped with the semiconductor die.

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